

Coupon Bight Aquatic Preserve

SEACAR Habitat Analyses

Last compiled on 10 June, 2026

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Funding & Acknowledgements

The data used in this analysis is from the Export Standardized Tables in the SEACAR Data Discovery Interface (DDI). Documents and information available through the SEACAR DDI are owned by the data provider(s) and users are expected to provide appropriate credit following accepted citation formats. Users are encouraged to access data to maximize utilization of gained knowledge, reducing redundant research and facilitating partnerships and scientific innovation.

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Threshold Filtering

Threshold filters, following the guidance of Florida Department of Environmental Protection's (*FDEP*) Division of Environmental Assessment and Restoration (*DEAR*) are used to exclude specific results values from the SEACAR Analysis. Based on the threshold filters, Quality Assurance / Quality Control (*QAQC*) Flags are inserted into the *SEACAR_QAQCFlagCode* and *SEACAR_QAQC_Description* columns of the export data. The *Include* column indicates whether the *QAQC* Flag will also indicate that data are excluded from analysis. No data are excluded from the data export, but the analysis scripts can use the *Include* column to exclude data (1 to include, 0 to exclude).

Table 1: Continuous Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Dissolved Oxygen	mg/L	-0.000001	50
Dissolved Oxygen Saturation	%	-0.000001	500
Fluorescent Dissolved Organic Matter	QSE	-	-
Salinity	ppt	-0.000001	70
Specific Conductivity	mS/cm	-0.000001	200
Turbidity	NTU	-0.000001	4000
Water Temperature	Degrees C	-5.000000	45
pH	None	2.000000	14

Table 2: Discrete Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Ammonia, Un-ionized (NH3)	mg/L	-	-
Ammonium (NH4)	mg/L	-	-

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Corrected for Pheophytin	ug/L	-	-
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Colored Dissolved Organic Matter	PCU	-	-
Dissolved Oxygen	mg/L	-0.000001	25
Dissolved Oxygen Saturation	%	-0.000001	310
Fluorescent Dissolved Organic Matter	QSE	-	-
Light Extinction Coefficient	m ⁻¹	-	-
NO2+3, Filtered	mg/L	-	-
Nitrate (NO3)	mg/L	-	-
Nitrite (NO2)	mg/L	-	-
Nitrogen, inorganic	mg/L	-	-
Nitrogen, organic	mg/L	-	-
Phosphate, Filtered (PO4)	mg/L	-	-
Salinity	ppt	-0.000001	70
Secchi Depth	m	0.000001	50
Specific Conductivity	mS/cm	0.005000	100
Total Ammonia (N)	mg/L	-	-
Total Kjeldahl Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Phosphorus	mg/L	-	-
Total Suspended Solids	mg/L	-	-
Turbidity	NTU	-	-
Water Temperature	Degrees C	3.000000	40
pH	None	2.000000	13

Table 3: Quality Assurance Flags inserted based on threshold checks listed in Table 1 and 2

<i>SEACAR QAQC Description</i>	<i>Include</i>	<i>SEACAR QAQCFlagCode</i>
Exceeds maximum threshold	0	2Q
Below minimum threshold	0	4Q
Within threshold tolerance	1	6Q
No defined thresholds for this parameter	1	7Q

Value Qualifiers

Value qualifier codes included within the data are used to exclude certain results from the analysis. The data are retained in the data export files, but the analysis uses the *Include* column to filter the results.

STORET and WIN value qualifier codes

Value qualifier codes from *STORET* and *WIN* data are examined with the database and used to populate the *Include* column in data exports.

Table 4: Value Qualifier codes excluded from analysis

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>MDL</i>	<i>Description</i>
STORET-WIN	H	0	0	Value based on field kit determination; results may not be accurate
STORET-WIN	J	0	0	Estimated value
STORET-WIN	V	0	0	Analyte was detected at or above method detection limit
STORET-WIN	Y	0	0	Lab analysis from an improperly preserved sample; data may be inaccurate

Discrete Water Quality Value Qualifiers

The following value qualifiers are highlighted in the Discrete Water Quality section of this report. An exception is made for **Program 476** - *Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network* and data flagged with Value Qualifier **H** are included for this program only.

H - Value based on field kit determination; results may not be accurate. This code shall be used if a field screening test (e.g., field gas chromatograph data, immunoassay, or vendor-supplied field kit) was used to generate the value and the field kit or method has not been recognized by the Department as equivalent to laboratory methods.

I - The reported value is greater than or equal to the laboratory method detection limit but less than the laboratory practical quantitation limit.

Q - Sample held beyond the accepted holding time. This code shall be used if the value is derived from a sample that was prepared or analyzed after the approved holding time restrictions for sample preparation or analysis.

S - Secchi disk visible to bottom of waterbody. The value reported is the depth of the waterbody at the location of the Secchi disk measurement.

U - Indicates that the compound was analyzed for but not detected. This symbol shall be used to indicate that the specified component was not detected. The value associated with the qualifier shall be the laboratory method detection limit. Unless requested by the client, less than the method detection limit values shall not be reported

Systemwide Monitoring Program (SWMP) value qualifier codes

Value qualifier codes from the *SWMP* continuous program are examined with the database and used to populate the *Include* column in data exports. *SWMP* Qualifier Codes are indicated by *QualifierSource=SWMP*.

Table 5: SWMP Value Qualifier codes

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>Description</i>
SWMP	-1	1	Optional parameter not collected
SWMP	-2	0	Missing data
SWMP	-3	0	Data rejected due to QA/QC
SWMP	-4	0	Outside low sensor range
SWMP	-5	0	Outside high sensor range
SWMP	0	1	Passed initial QA/QC checks
SWMP	1	0	Suspect data
SWMP	2	1	Reserved for future use
SWMP	3	1	Calculated data: non-vented depth/level sensor correction for changes in barometric pressure
SWMP	4	1	Historical: Pre-auto QA/QC
SWMP	5	1	Corrected data

Water Column

The water column habitat extends from the water's surface to the bottom sediments, and it's where fish, dolphins, crabs and people swim! So much life makes its home in the water column that the health of marine and coastal ecosystems, as well as human economies, depend on the condition of this vulnerable habitat. Local patterns of rainfall, temperature, winds and currents can rapidly change the condition of the water column, while global influences such as [El Niño/La Niña](#), large-scale fluctuation in sea temperatures and climate change can have long-term effects. Inputs from the prosperity of our day-to-day lives including farming, mining and forestry, and emissions from power generation, automobiles and water treatment can also alter the health of the water column. Acting alone or together, each input can have complex and lasting effects on habitats and ecosystems.

SEACAR evaluates water column health with several essential parameters. These include nutrient surveys of nitrogen and phosphorus, and water quality assessments of salinity, dissolved oxygen, pH, and water temperature. Water clarity is evaluated with Secchi depth, turbidity, levels of chlorophyll a, total suspended solids, and colored dissolved organic matter. Additionally, the richness of nekton is indicated by the abundance of free-swimming fishes and macroinvertebrates like crabs and shrimps.

Seasonal Kendall-Tau Analysis

Indicators must have a minimum of five to ten years, depending on the habitat, of data within the geographic range of the analysis to be included in the analysis. Ten years of data are required for discrete parameters, and five years of data are required for continuous parameters. If there are insufficient years of data, the number of years of data available will be noted and labeled as "insufficient data to conduct analysis". Further, for the preferred Seasonal Kendall-Tau test, there must be data from at least two months in common across at least two consecutive years within the RCP managed area being analyzed. Values that pass both of these tests will be included in the analysis and be labeled as *Use_In_Analysis* = **TRUE**. Any that fail either test will be excluded from the analyses and labeled as *Use_In_Analysis* = **FALSE**. The points for all Water Column plots displayed in this section are monthly averages. Trend significance will be denoted as "Significant Trend" (when $p < 0.05$), or "Non-significant Trend" (when $p \geq 0.05$). Any parameters with insufficient data to perform Seasonal Kendall-Tau test will have their monthly averages plotted without a corresponding trend line.

Water Quality - Discrete

The following files were used in the discrete analysis:

- *Combined_WQ_WC_NUT_Chlorophyll_a_corrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Chlorophyll_a_uncorrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Colored_dissolved_organic_matter_CDOM-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen_Saturation-2026-May-18.txt*
- *Combined_WQ_WC_NUT_pH-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Salinity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Secchi_Depth-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Nitrogen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Phosphorus-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Suspended_Solids_TSS-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Turbidity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Water_Temperature-2026-May-18.txt*

Chlorophyll a, Uncorrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

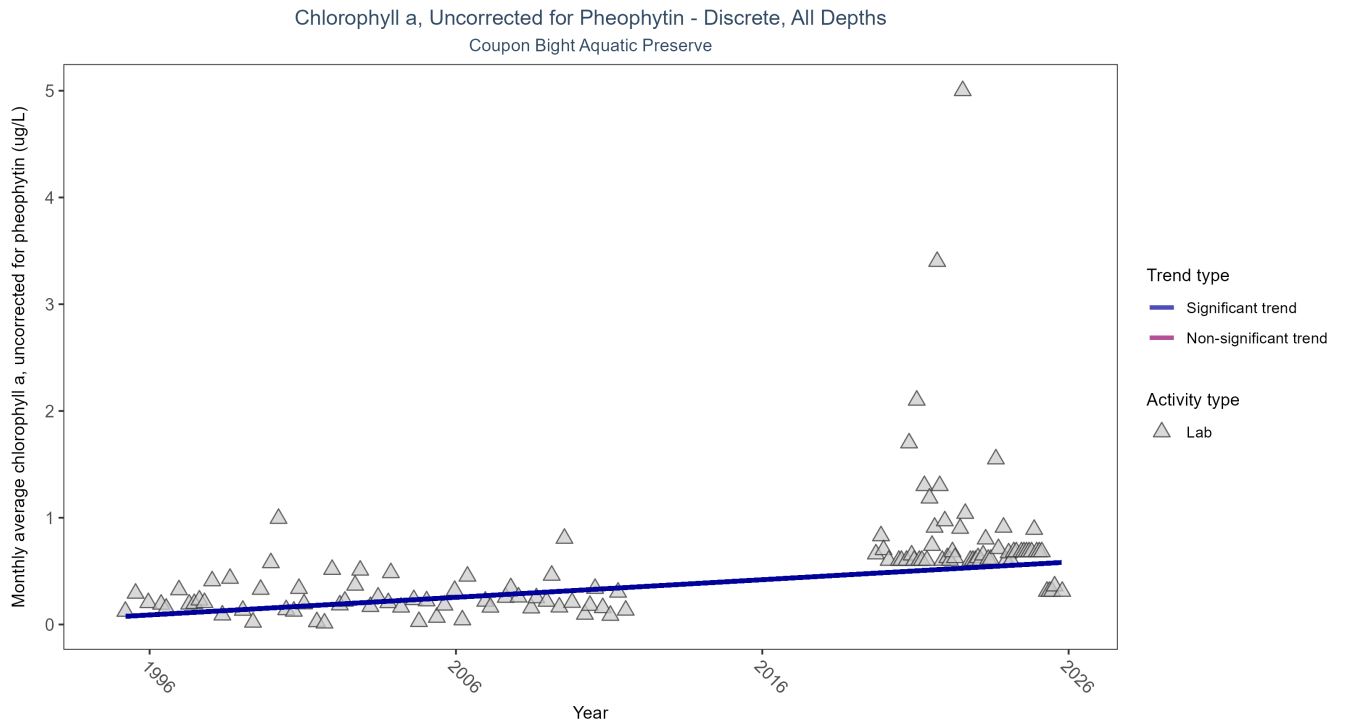


Figure 1: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 6: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	173	24	1995 - 2025	0.6	0.4357	0.073	0.0165	0

Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.02 $\mu\text{g/L}$ per year, indicating a decrease in water clarity.

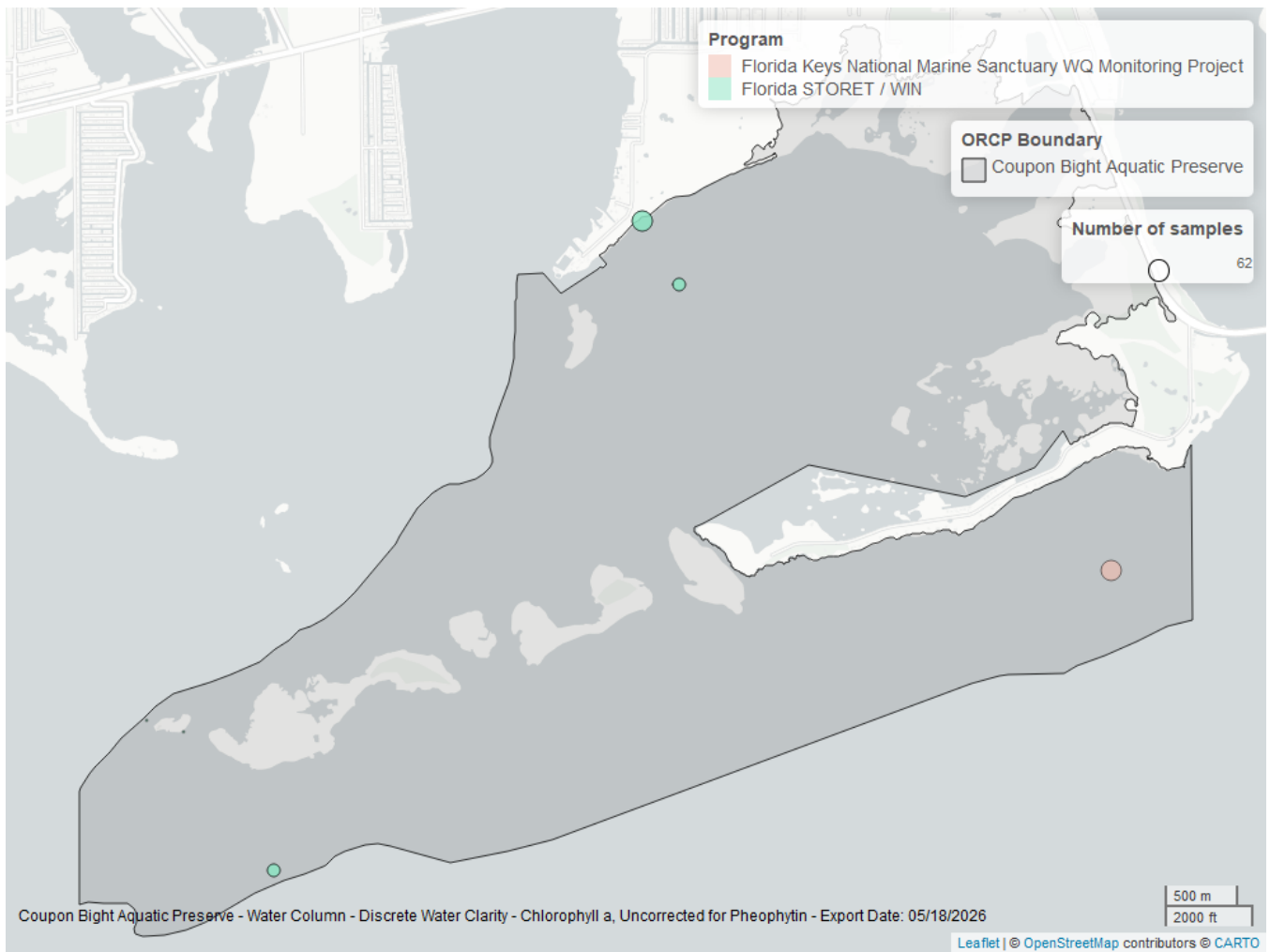


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 7: Programs contributing data for Chlorophyll a, Uncorrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
5002	114	2019	2025
297	61	1995	2011

Program names:

5002 - Florida STORET / WIN¹

297 - Florida Keys National Marine Sanctuary Water Quality Monitoring Project²

Dissolved Oxygen - Discrete

Seasonal Kendall-Tau Trend Analysis

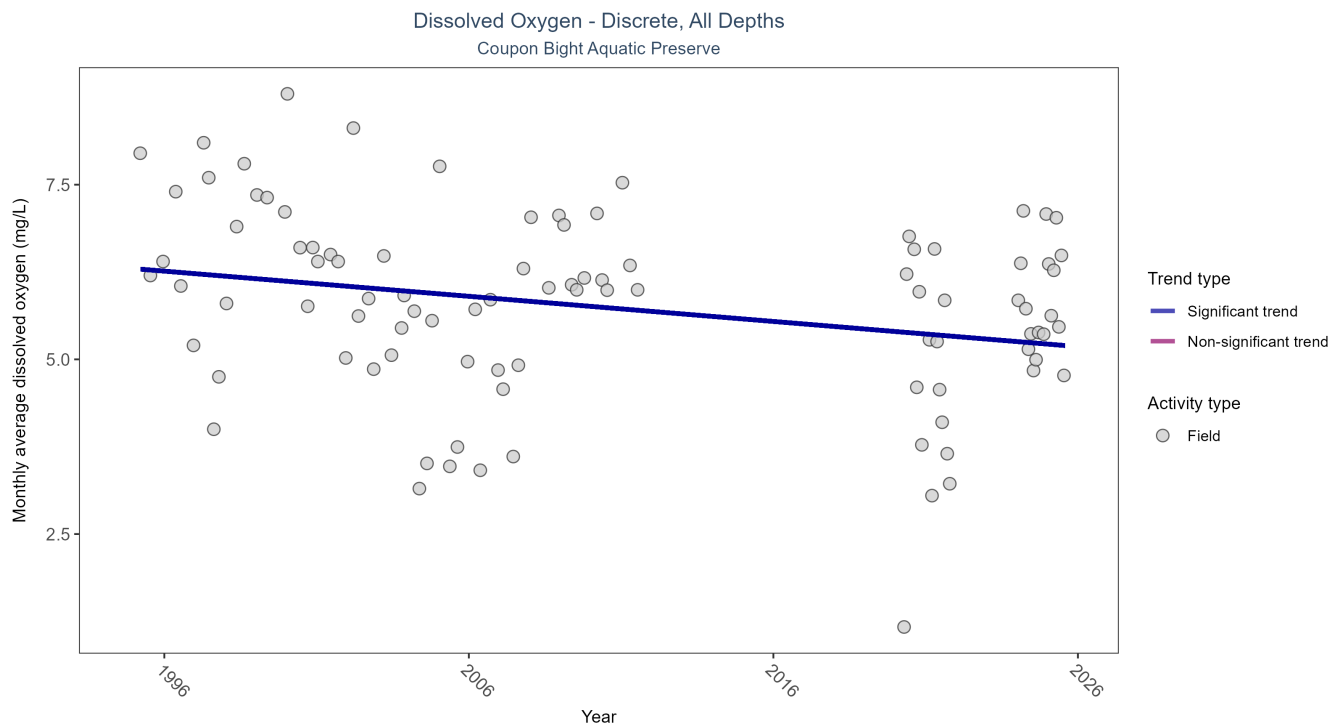


Figure 3: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 8: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	195	21	1995 - 2025	5.9901	-0.1974	6.2985	-0.036	0.0175

Monthly average dissolved oxygen decreased by 0.04 mg/L per year.

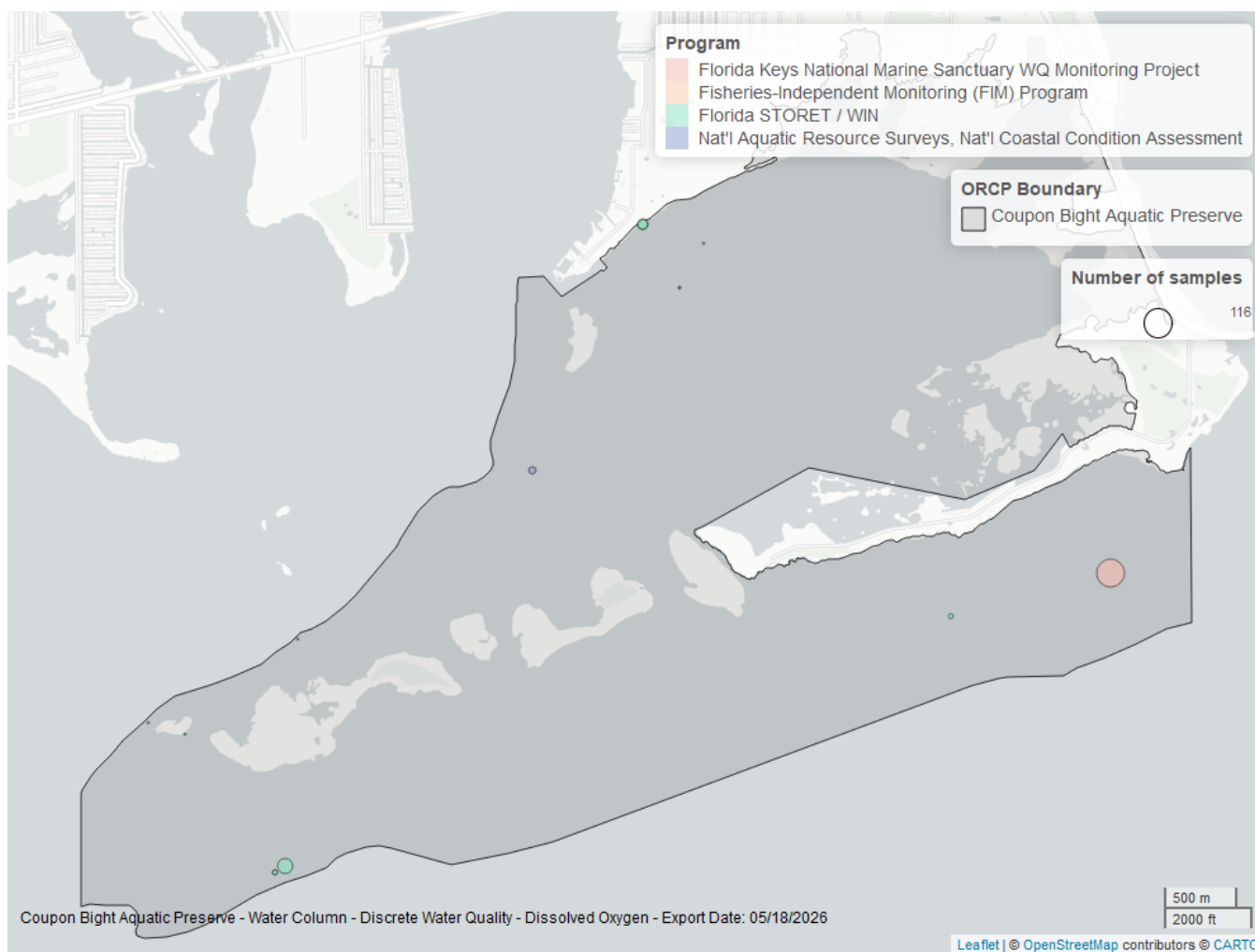


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 9: Programs contributing data for Dissolved Oxygen

ProgramID	N_Data	YearMin	YearMax
297	116	1995	2011
5002	68	2020	2025
118	7	2020	2020
69	4	2000	2000

Program names:

297 - Florida Keys National Marine Sanctuary Water Quality Monitoring Project²

5002 - Florida STORET / WIN¹

118 - National Aquatic Resource Surveys, National Coastal Condition Assessment³

69 - Fisheries-Independent Monitoring (FIM) Program⁴

Dissolved Oxygen Saturation - Discrete

Seasonal Kendall-Tau Trend Analysis

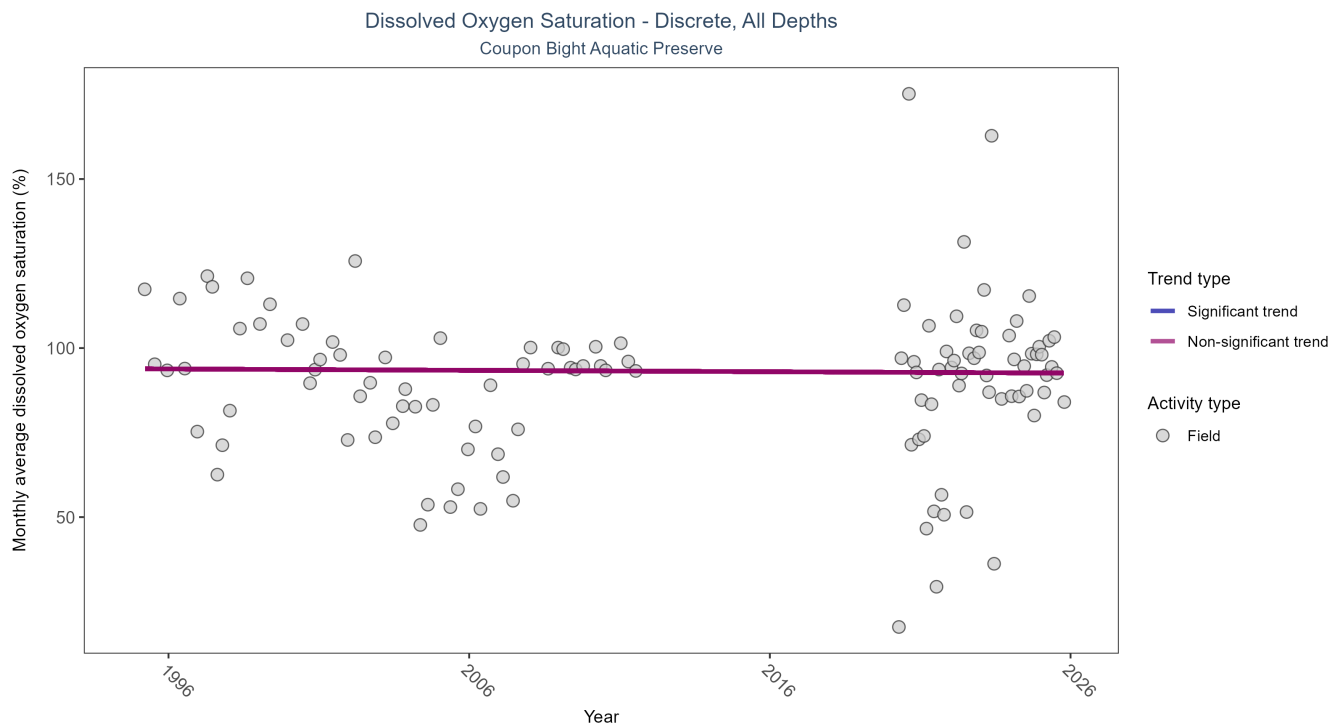


Figure 5: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 10: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen Saturation

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	245	23	1995 - 2025	93.6199	-0.0116	93.866	-0.041	0.8392

Dissolved oxygen saturation showed no detectable trend between 1995 and 2025.

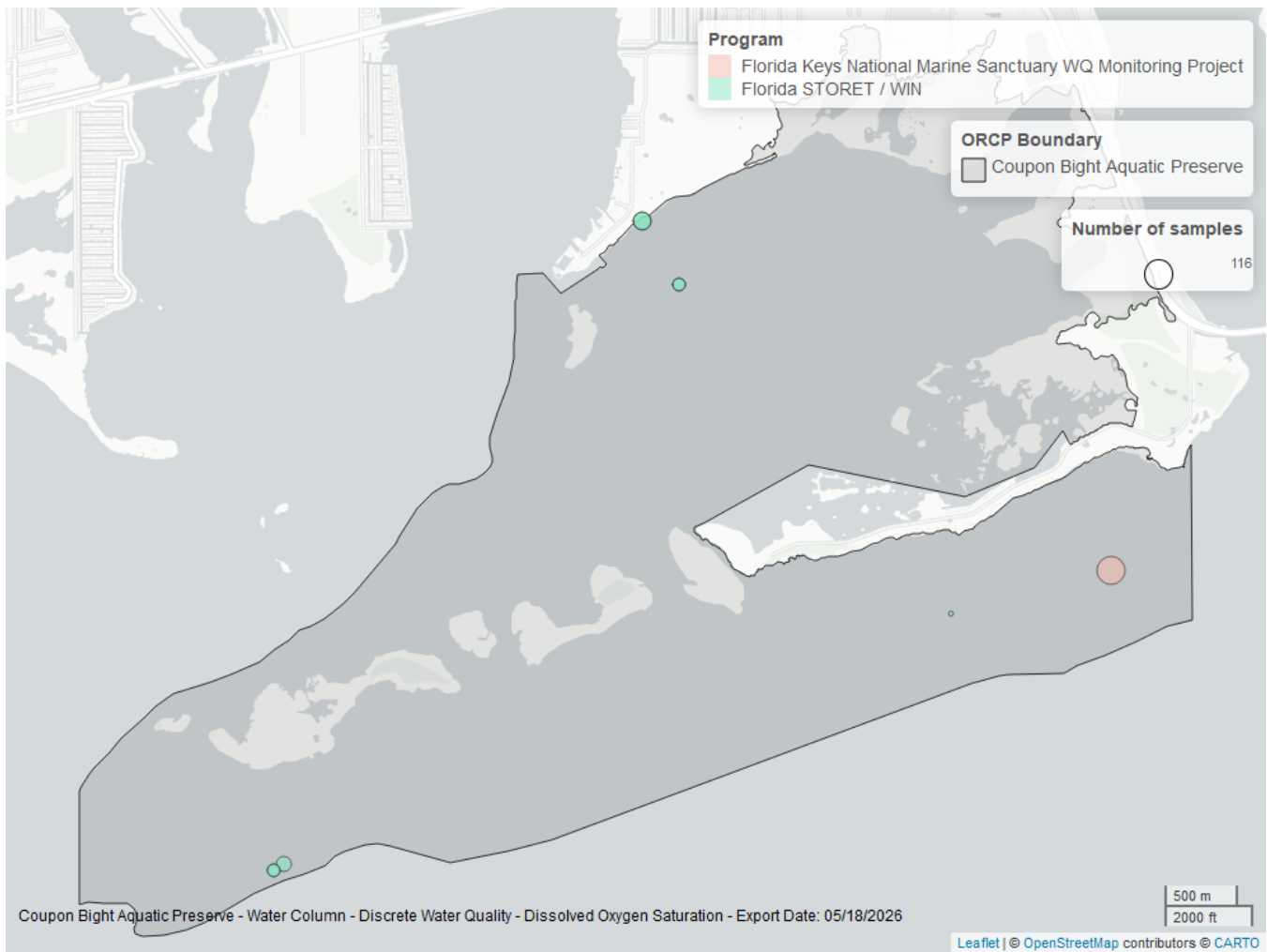


Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 11: Programs contributing data for Dissolved Oxygen Saturation

ProgramID	N_Data	YearMin	YearMax
5002	134	2020	2025
297	116	1995	2011

Program names:

5002 - Florida STORET / WIN¹

297 - Florida Keys National Marine Sanctuary Water Quality Monitoring Project²

Salinity - Discrete

Seasonal Kendall-Tau Trend Analysis

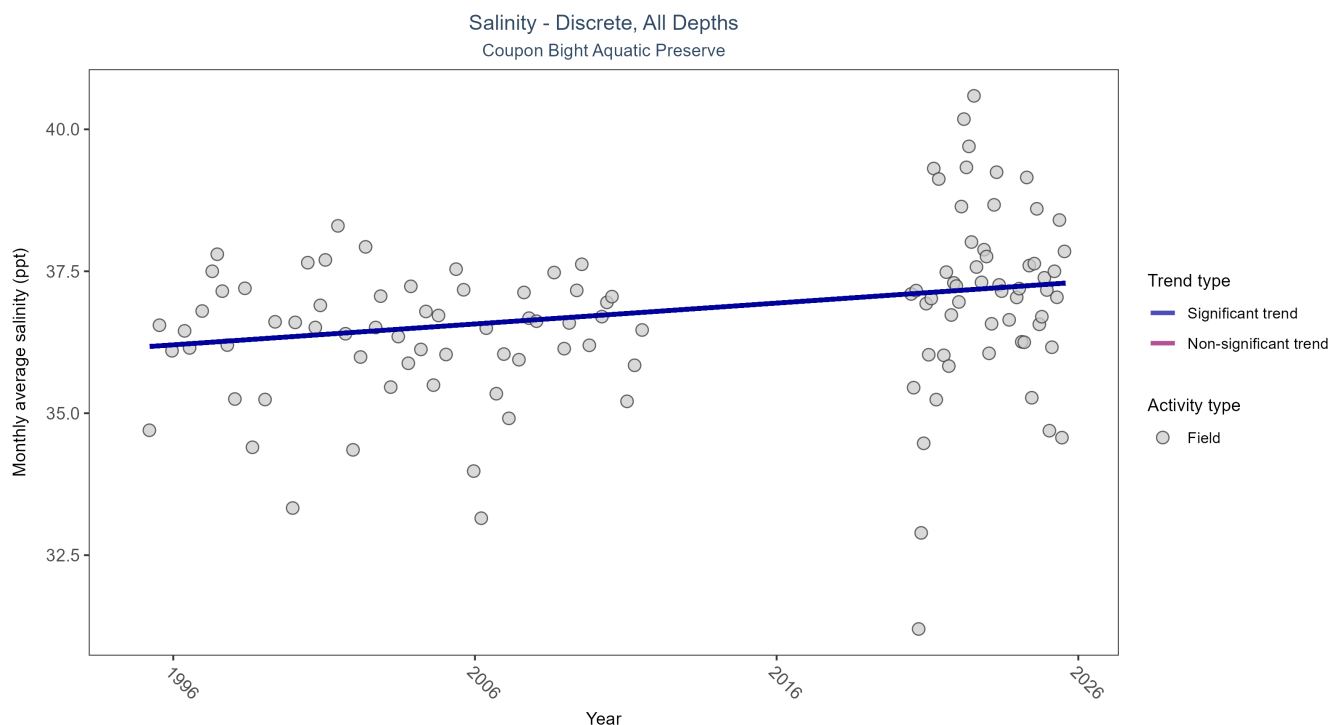


Figure 7: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 12: Seasonal Kendall-Tau Trend Analysis for Salinity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Increasing trend	249	23	1995 - 2025	36.7483	0.269	36.1675	0.0368	0.0001

Monthly average salinity increased by 0.04 ppt per year.

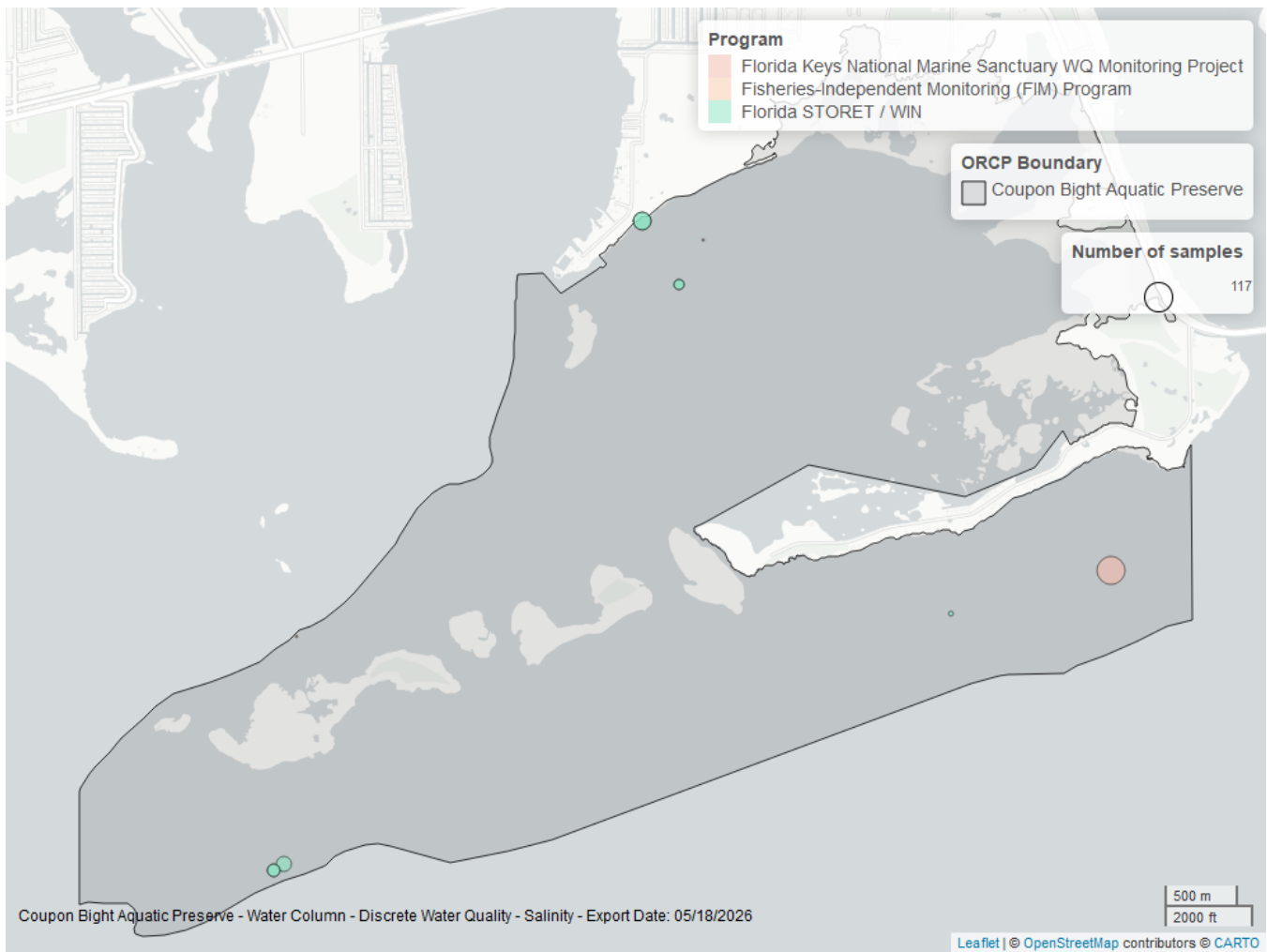


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 13: Programs contributing data for Salinity

ProgramID	N_Data	YearMin	YearMax
5002	133	2020	2025
297	117	1995	2011
69	4	2000	2000

Program names:

5002 - Florida STORET / WIN¹

297 - Florida Keys National Marine Sanctuary Water Quality Monitoring Project²

69 - Fisheries-Independent Monitoring (FIM) Program⁴

Total Nitrogen - Discrete

Total Nitrogen Calculation:

The logic for calculated Total Nitrogen was provided by Kevin O'Donnell and colleagues at FDEP (with the help of Jay Silvanima, Watershed Monitoring Section). The following logic is used, in this order, based on the availability of specific nitrogen components.

- 1) $TN = TKN + NO3O2$;
- 2) $TN = TKN + NO3 + NO2$;
- 3) $TN = ORGN + NH4 + NO3O2$;
- 4) $TN = ORGN + NH4 + NO2 + NO3$;
- 5) $TN = TKN + NO3$;
- 6) $TN = ORGN + NH4 + NO3$;

Additional Information:

- Rules for use of sample fraction:
 - Florida Department of Environmental Protection (FDEP) report that if both “Total” and “Dissolved” components are reported, only “Total” is used. If the total is not reported, then the dissolved components are used as a best available replacement.
 - Total nitrogen calculations are done using nitrogen components with the same sample fraction, nitrogen components with mixed total/dissolved sample fractions are not used. In other words, total nitrogen can be calculated when TKN and NO3O2 are both total sample fractions, or when both are dissolved sample fractions. *Future calculations of total nitrogen values may be based on components with mixed sample fractions.*
- Values inserted into data:
 - ParameterName = “Total Nitrogen”
 - SEACAR_QAQCFlagCode = “1Q”
 - SEACAR_QAQC_Description = “SEACAR Calculated”

Seasonal Kendall-Tau Trend Analysis

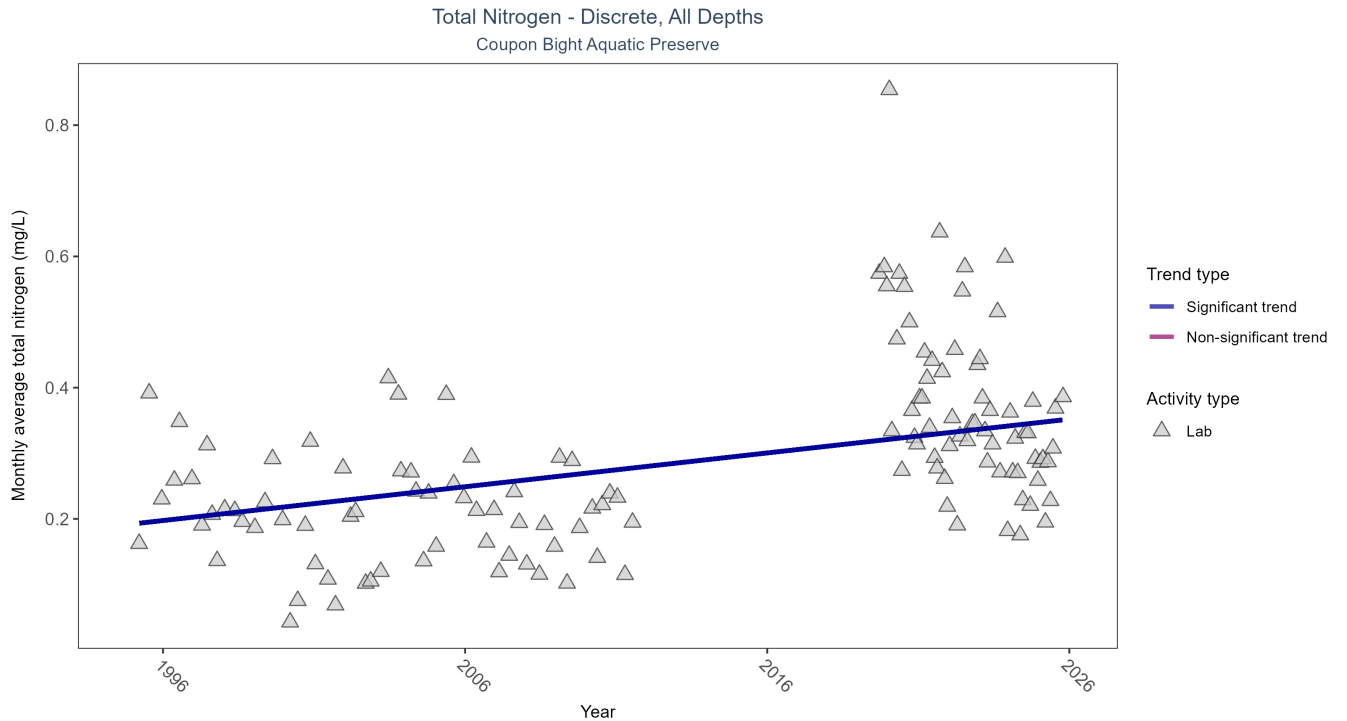


Figure 9: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 14: Seasonal Kendall-Tau Trend Analysis for Total Nitrogen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	219	24	1995 - 2025	0.264	0.2752	0.1923	0.0052	0.0001

Monthly average total nitrogen increased by less than 0.01 mg/L per year.

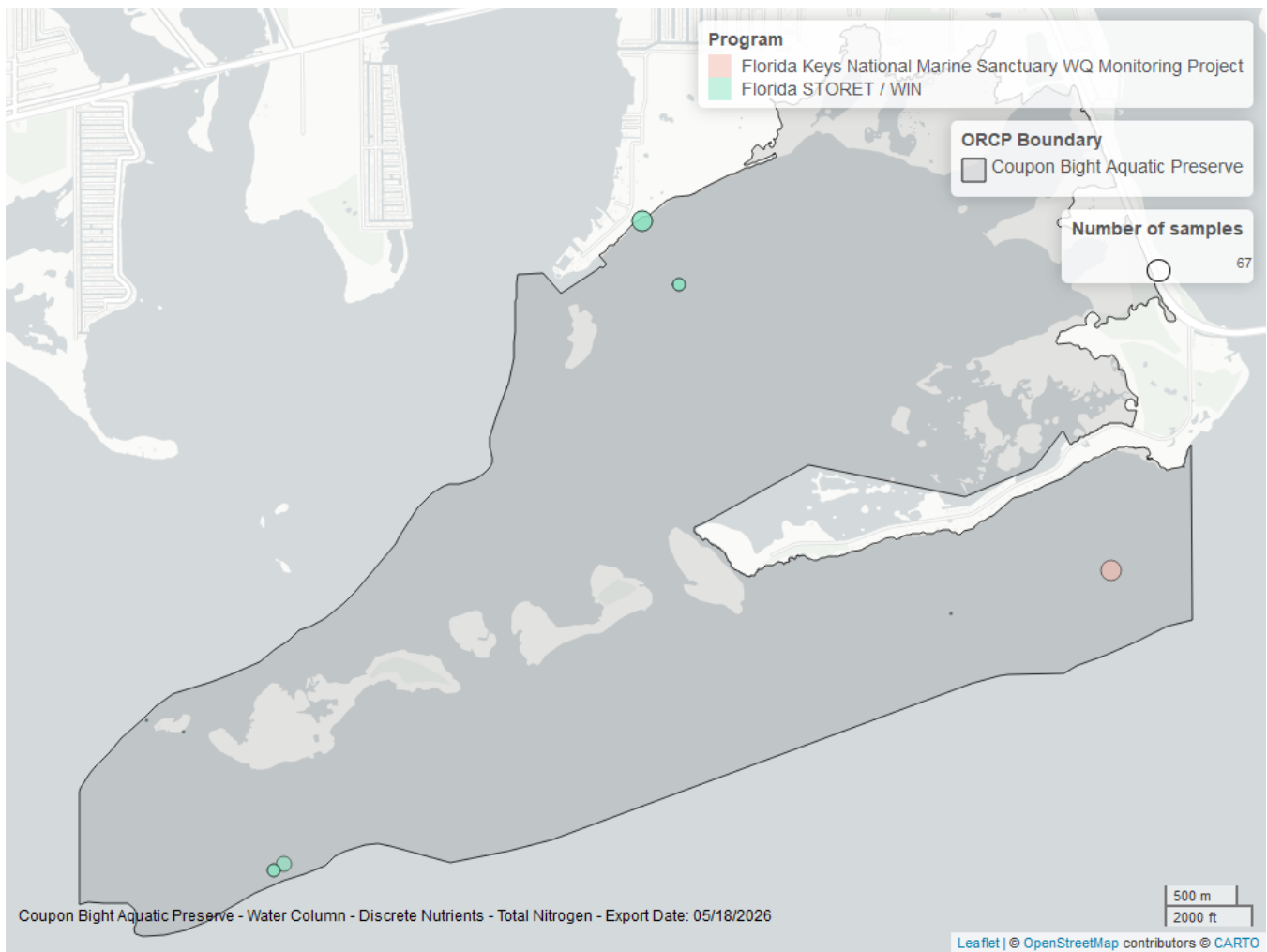


Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 15: Programs contributing data for Total Nitrogen

ProgramID	N_Data	YearMin	YearMax
5002	152	2019	2025
297	67	1995	2011

Program names:

5002 - Florida STORET / WIN¹

297 - Florida Keys National Marine Sanctuary Water Quality Monitoring Project²

Total Phosphorus - Discrete

Seasonal Kendall-Tau Trend Analysis

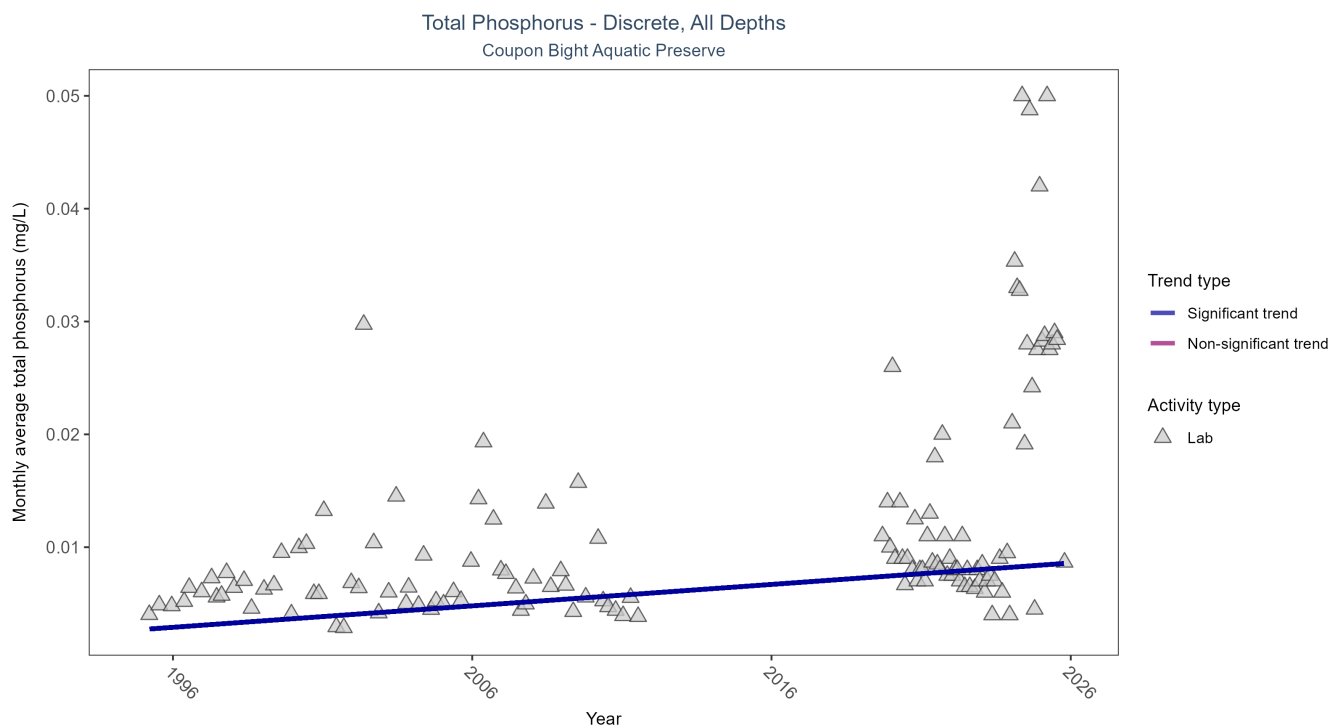


Figure 11: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 16: Seasonal Kendall-Tau Trend Analysis for Total Phosphorus

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	217	24	1995 - 2025	0.008	0.3854	0.0027	0.0002	0

Monthly average total phosphorus increased by less than 0.01 mg/L per year.

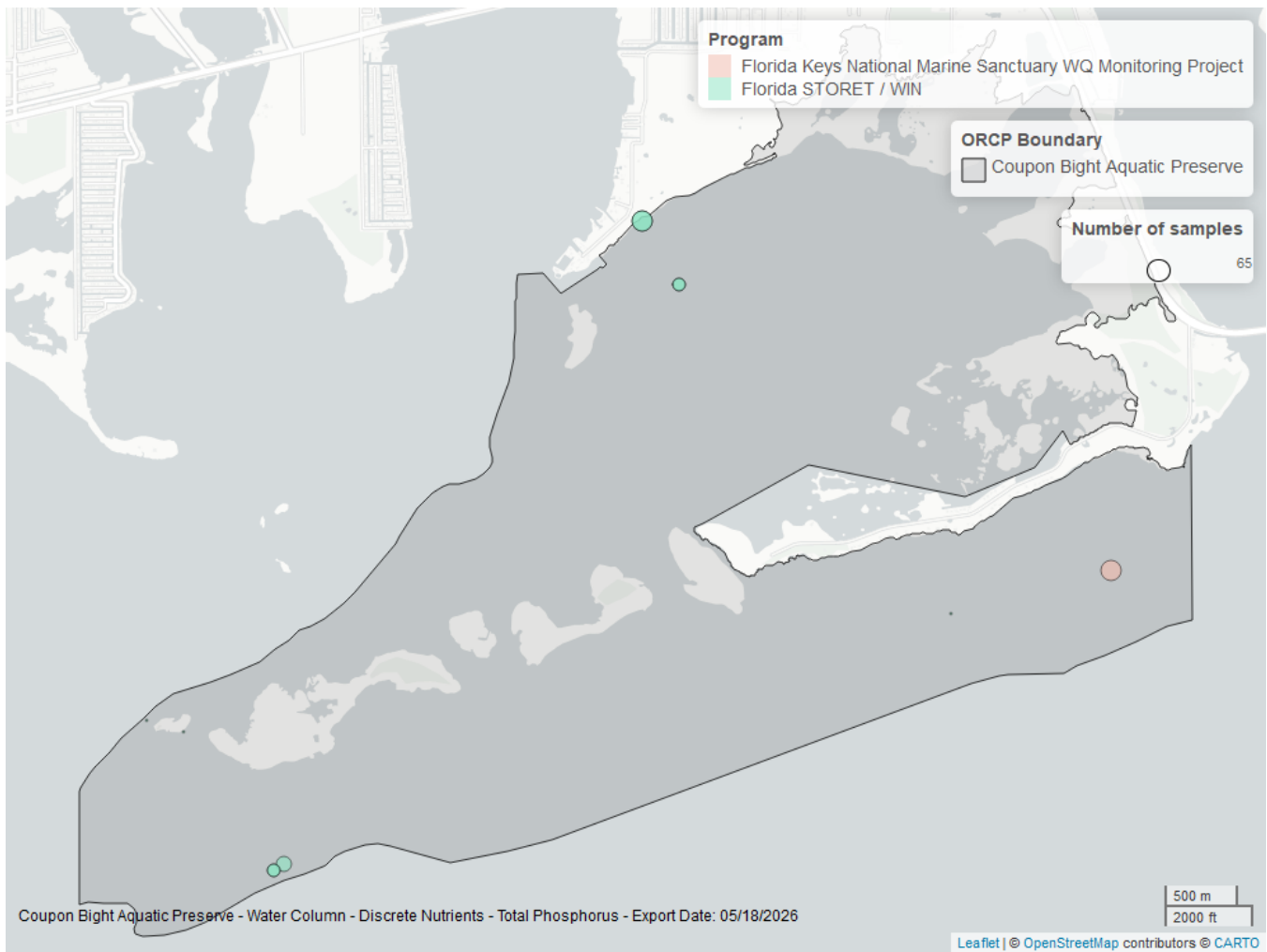


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 17: Programs contributing data for Total Phosphorus

ProgramID	N_Data	YearMin	YearMax
5002	155	2019	2025
297	64	1995	2011

Program names:

5002 - Florida STORET / WIN¹

297 - Florida Keys National Marine Sanctuary Water Quality Monitoring Project²

Turbidity - Discrete

Seasonal Kendall-Tau Trend Analysis

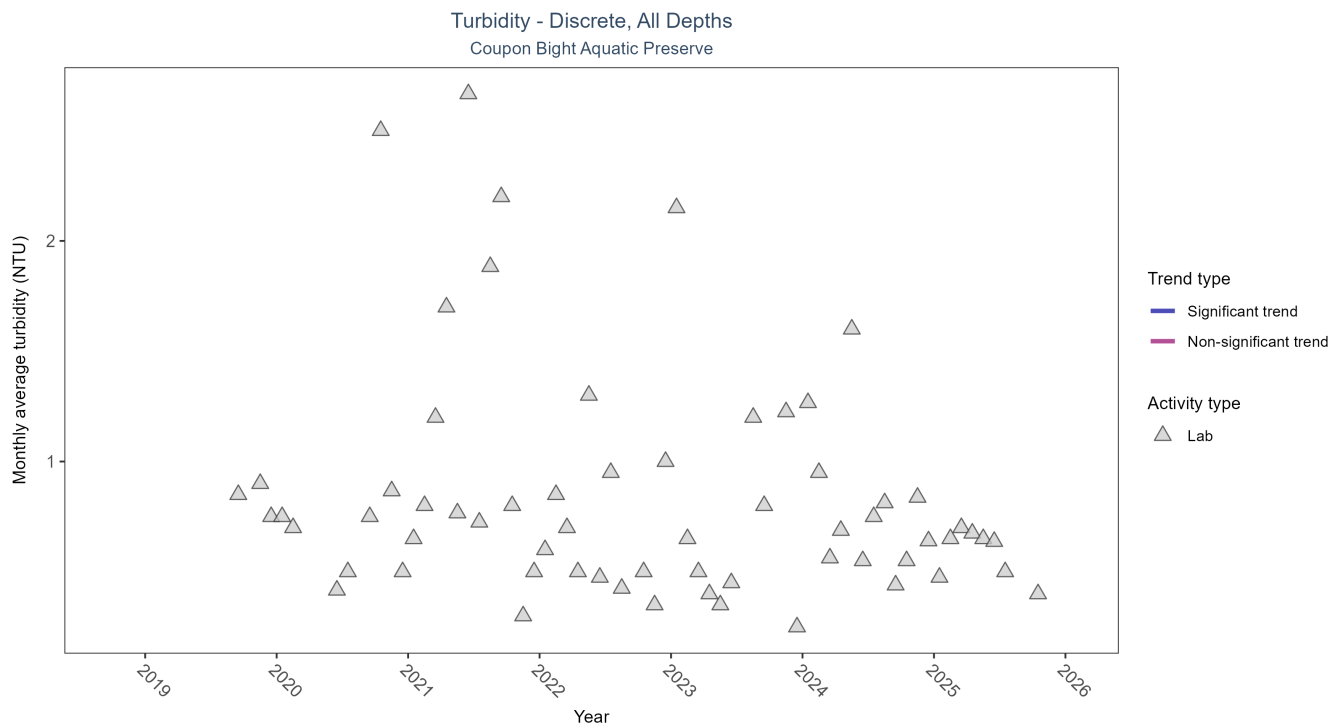


Figure 13: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 18: Seasonal Kendall-Tau Trend Analysis for Turbidity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data	148	7	2019 - 2025	0.65	-	-	-	-

There was insufficient data to fit a model for turbidity.

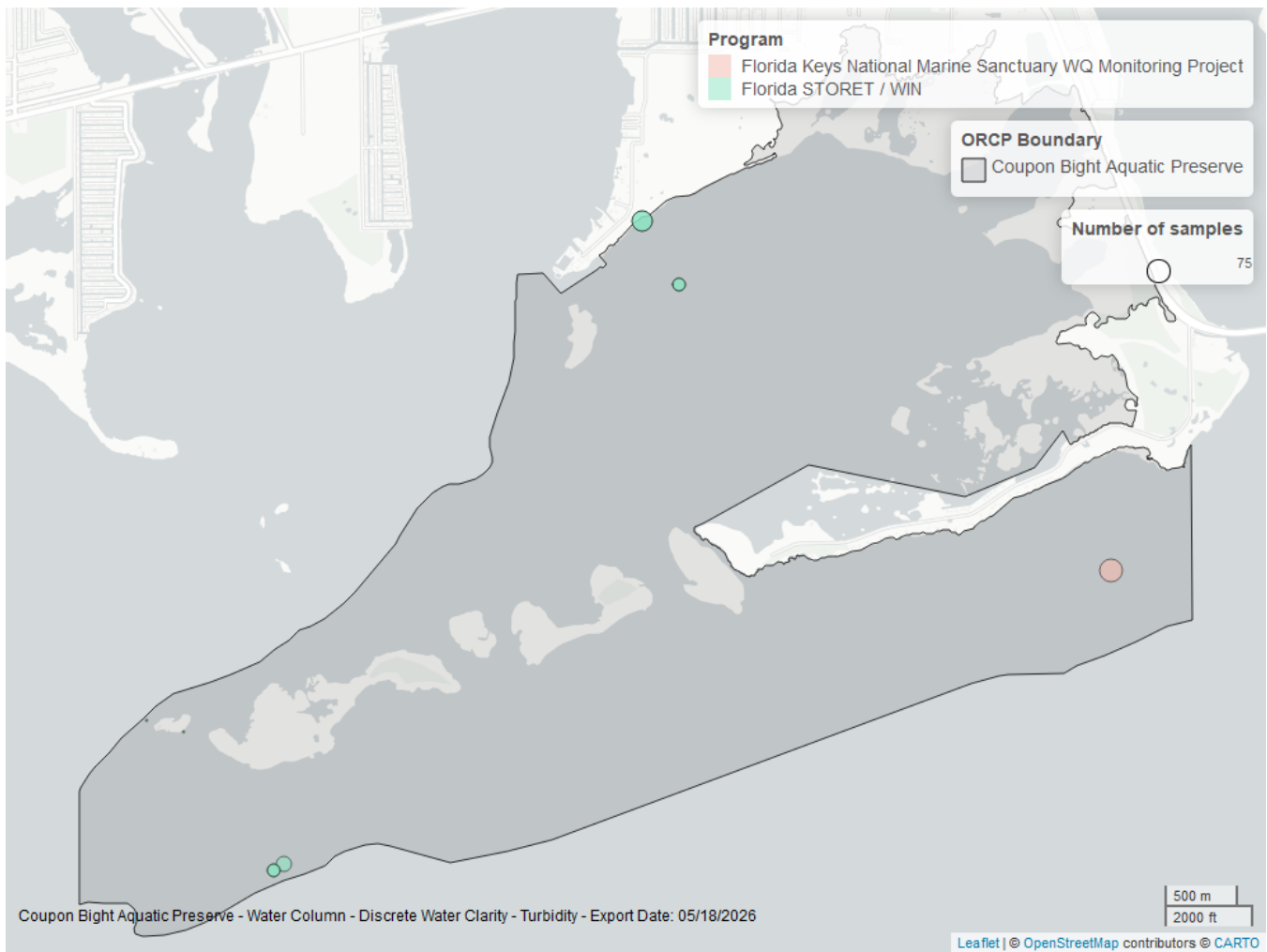


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 19: Programs contributing data for Turbidity

ProgramID	N_Data	YearMin	YearMax
5002	156	2019	2025
297	75	1995	2011

Program names:

5002 - Florida STORET / WIN¹

297 - Florida Keys National Marine Sanctuary Water Quality Monitoring Project²

Water Temperature - Discrete

Seasonal Kendall-Tau Trend Analysis

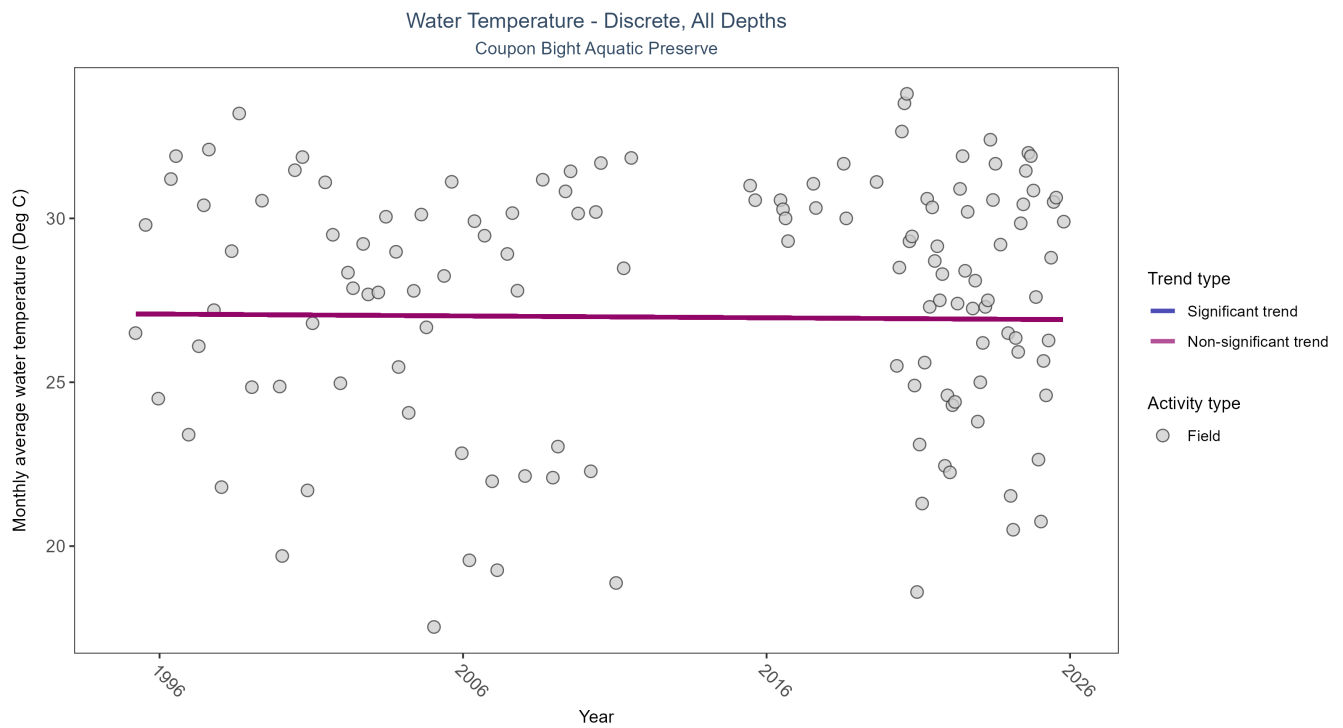


Figure 15: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 20: Seasonal Kendall-Tau Trend Analysis for Water Temperature

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	290	28	1995 - 2025	28.55	-0.0161	27.0863	-0.0057	0.6083

Water temperature showed no detectable trend between 1995 and 2025.

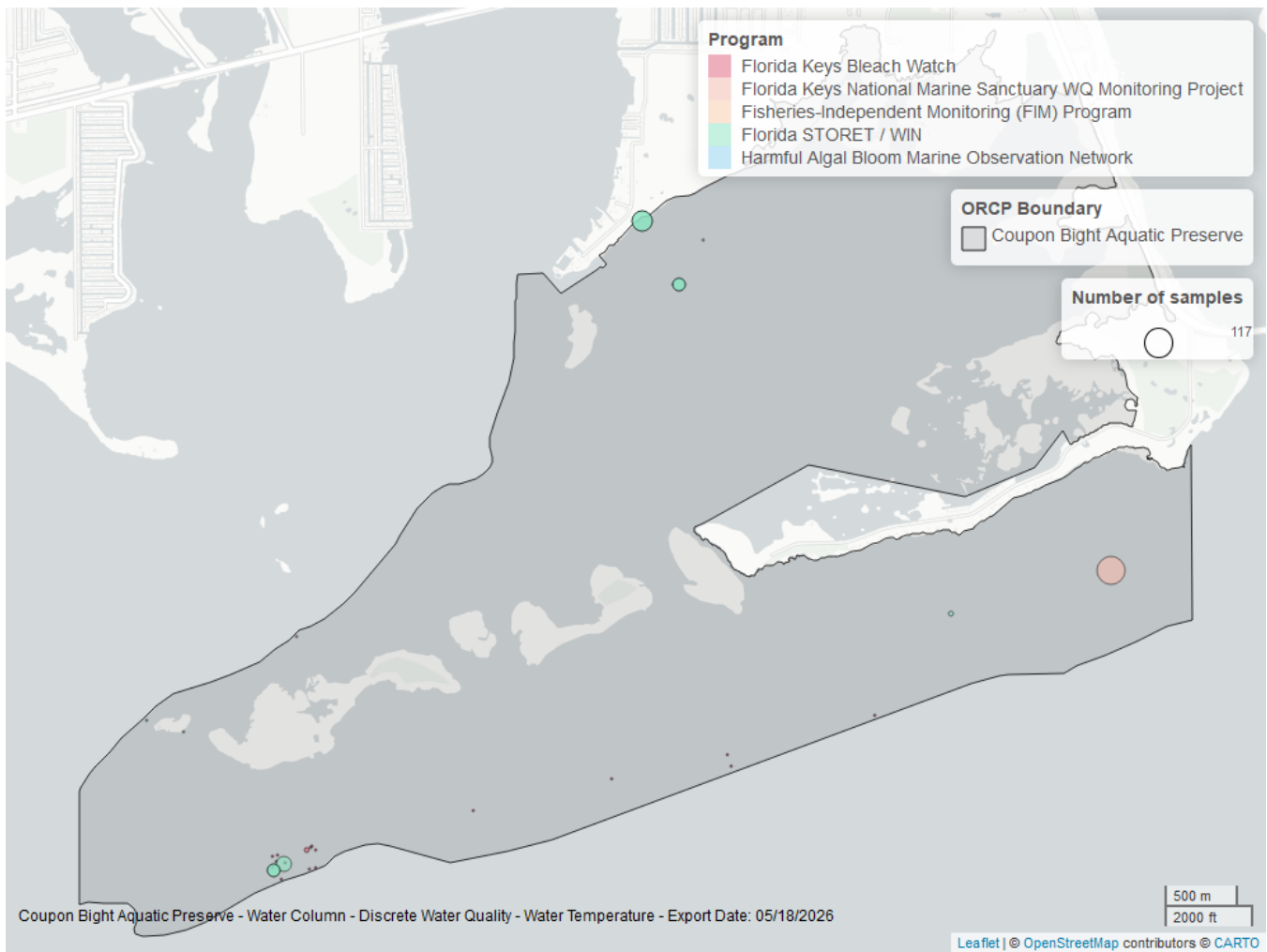


Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 21: Programs contributing data for Water Temperature

ProgramID	N_Data	YearMin	YearMax
5002	152	2020	2025
297	117	1995	2011
982	23	2015	2023
69	4	2000	2000
95	1	2010	2010

Program names:

5002 - Florida STORET / WIN¹

297 - Florida Keys National Marine Sanctuary Water Quality Monitoring Project²

982 - Florida Keys Bleach Watch⁵

69 - Fisheries-Independent Monitoring (FIM) Program⁴

95 - Harmful Algal Bloom Marine Observation Network⁶

Water Quality - Continuous

The following files were used in the continuous analysis:

- *Combined_WQ_WC_NUT_cont_Chlorophyll_a_Uncorrected_for_Pheophytin-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen_Saturation-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Fluorescent_Dissolved_Organic_Matter-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_pH-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Salinity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Specific_Conductivity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Turbidity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Water_Temperature-2026-Mar-06.txt*

Continuous monitoring locations in Coupon Bight Aquatic Preserve

Table 22: Station overview for Continuous parameters by Program

<i>ProgramID</i>	<i>ProgramLocationID</i>	<i>Years of Data</i>	<i>Use in Analysis</i>	<i>Parameters</i>
10004	FKCB	1	FALSE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity

Program names:

[10004](#) - Florida Keys Aquatic Preserves Continuous Water Quality Monitoring⁷

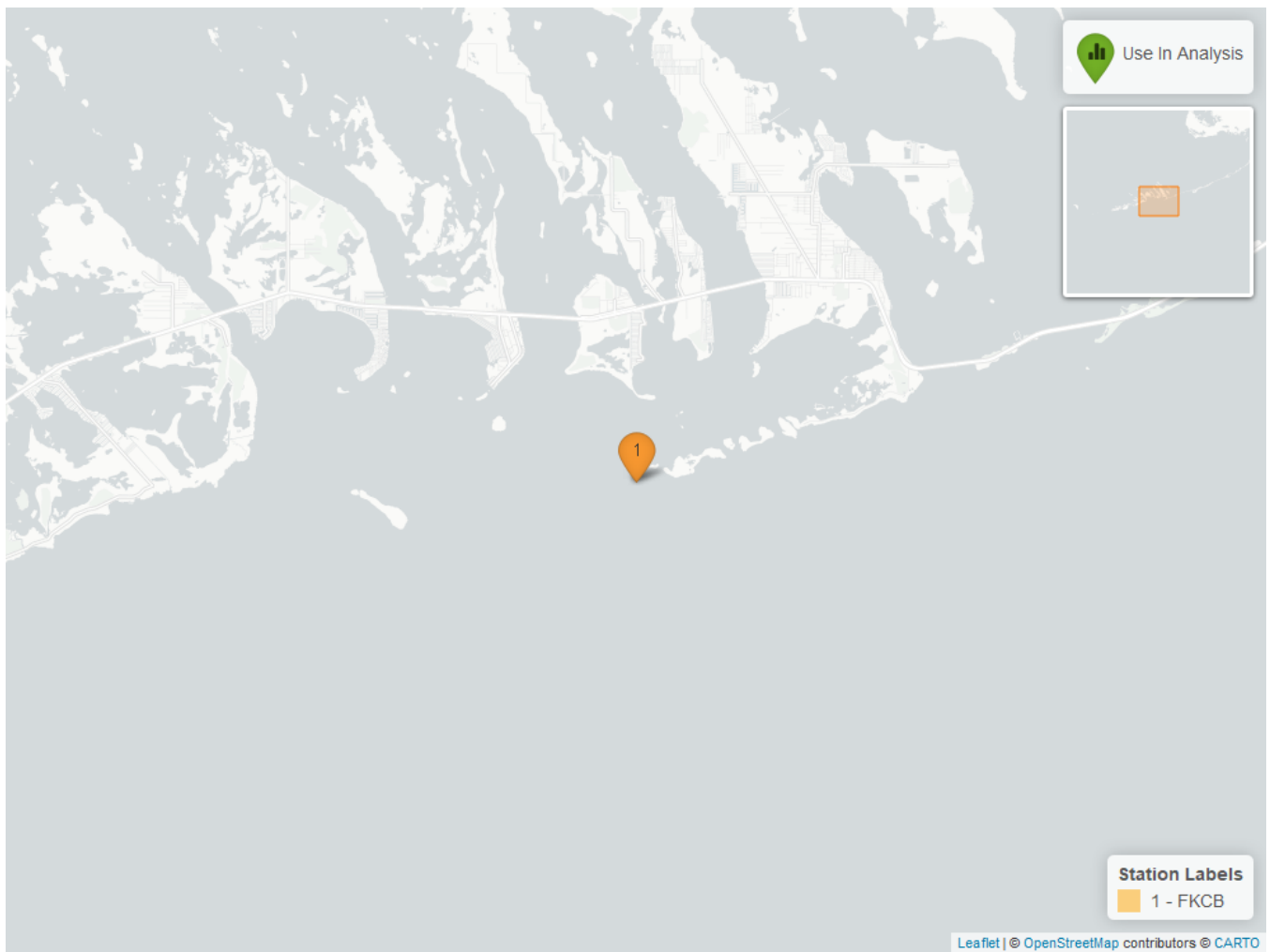


Figure 17: Map showing continuous water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. Sites marked as *Use In Analysis* (green) are featured in this report.

pH - Continuous

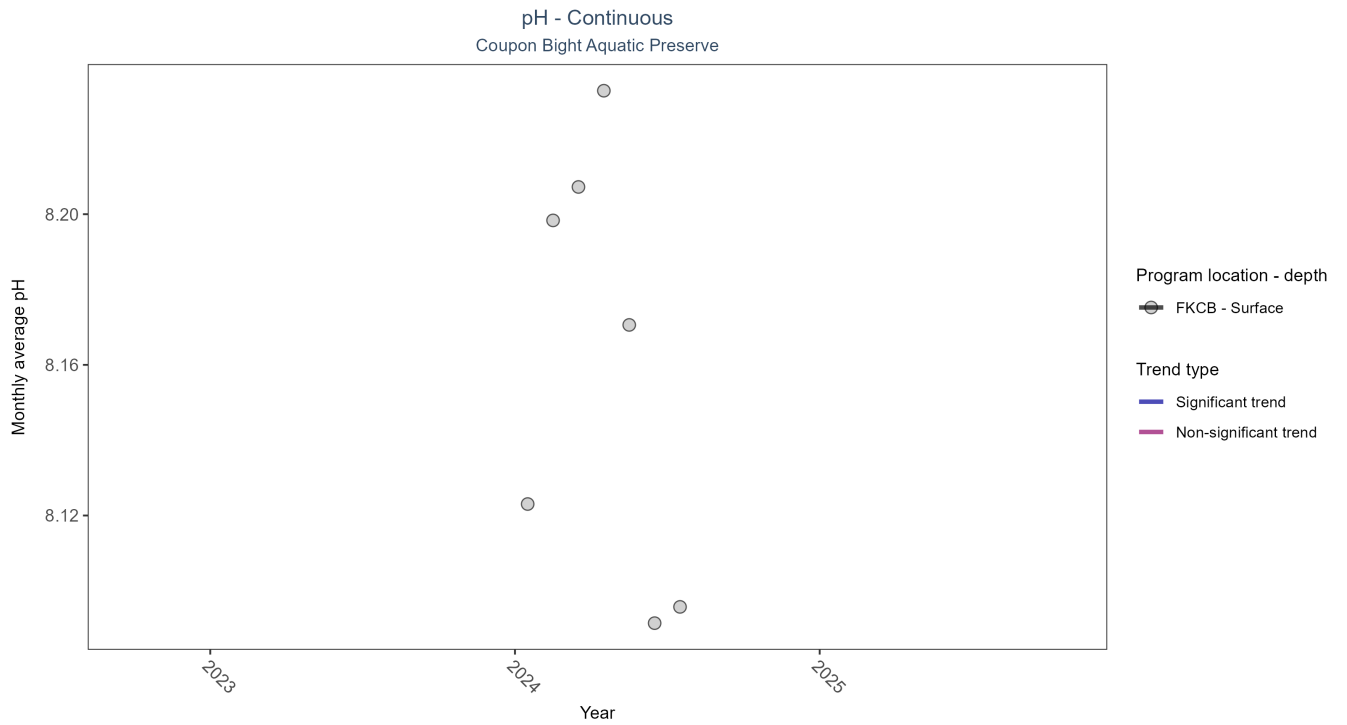


Figure 18: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 23: Seasonal Kendall-Tau Results for pH - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data to calculate trend	16263	1	2024 - 2024	8.2	-	-	-	-

There was insufficient data to fit a model for one location.

Dissolved Oxygen Saturation - Continuous

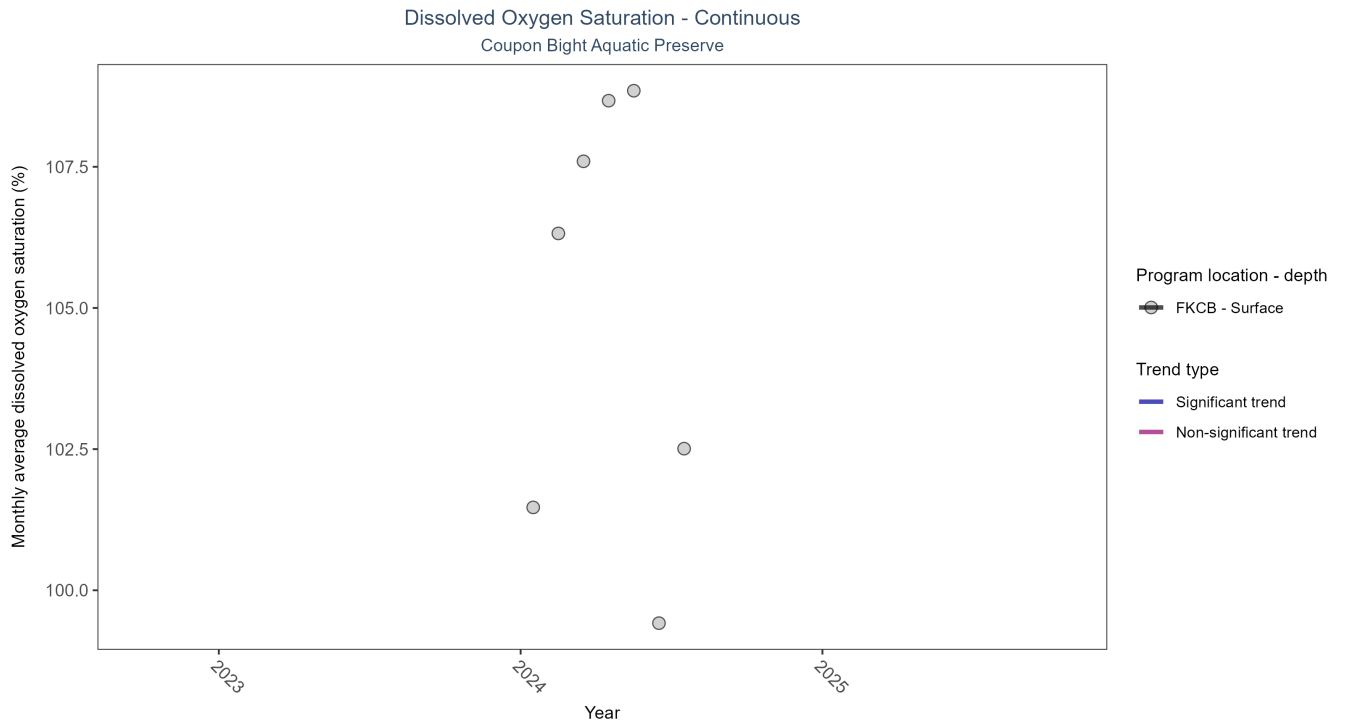


Figure 19: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 24: Seasonal Kendall-Tau Results for Dissolved Oxygen Saturation - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data to calculate trend	16263	1	2024 - 2024	103.3	-	-	-	-

There was insufficient data to fit a model for one location.

Dissolved Oxygen - Continuous

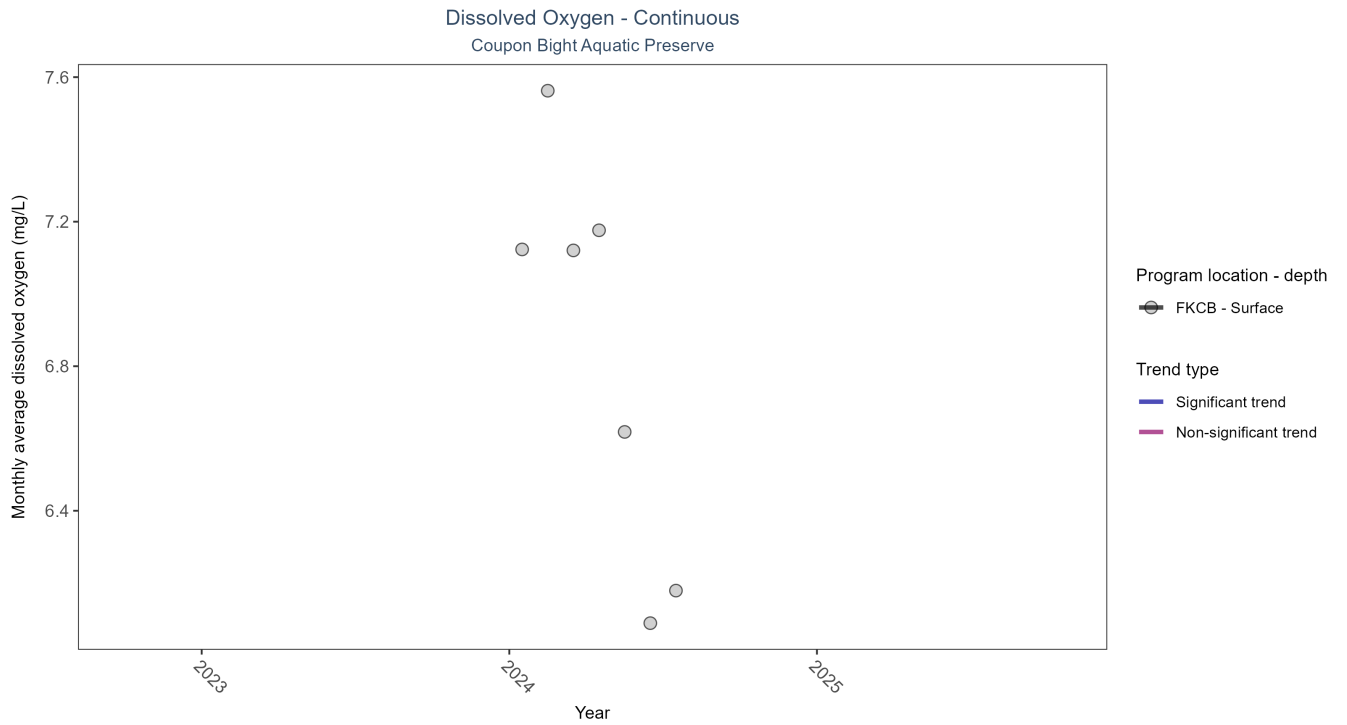


Figure 20: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 25: Seasonal Kendall-Tau Results for Dissolved Oxygen - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data to calculate trend	16262	1	2024 - 2024	6.8	-	-	-	-

There was insufficient data to fit a model for one location.

Turbidity - Continuous

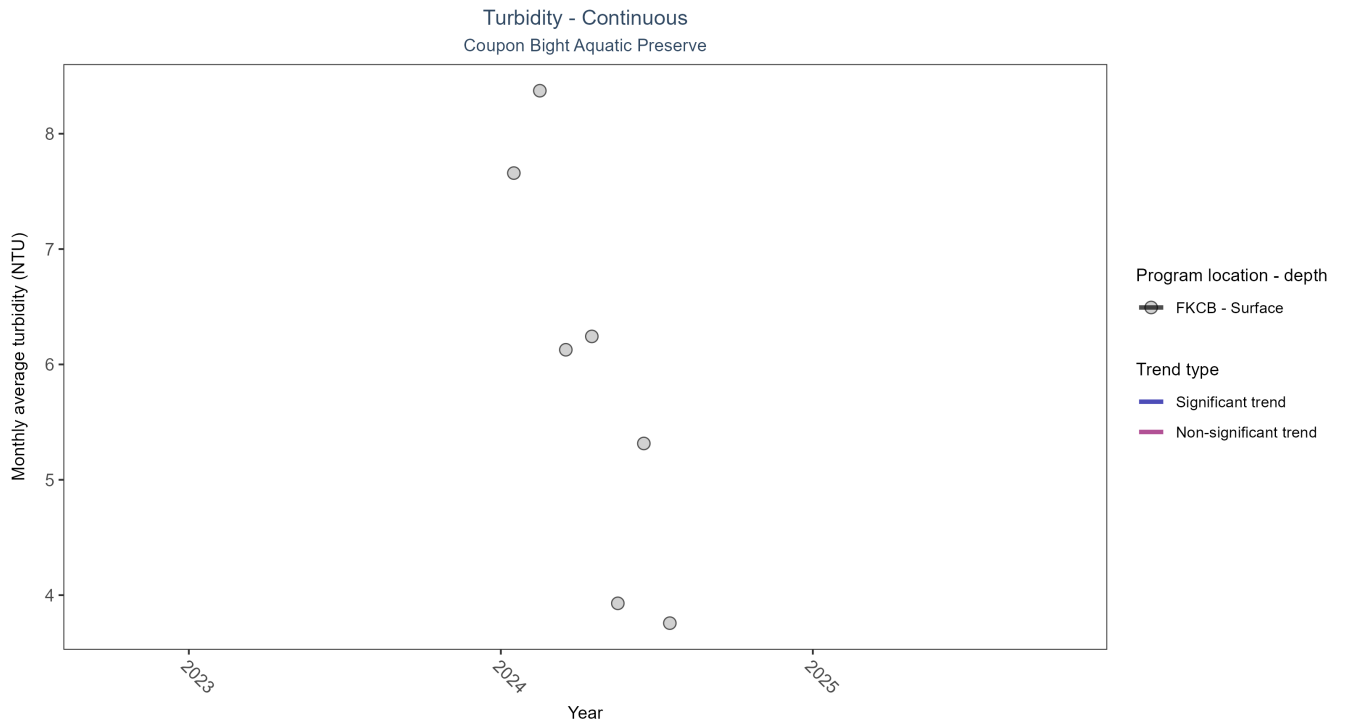


Figure 21: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 26: Seasonal Kendall-Tau Results for Turbidity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data to calculate trend	16240	1	2024 - 2024	4	-	-	-	-

There was insufficient data to fit a model for one location.

Water Temperature - Continuous

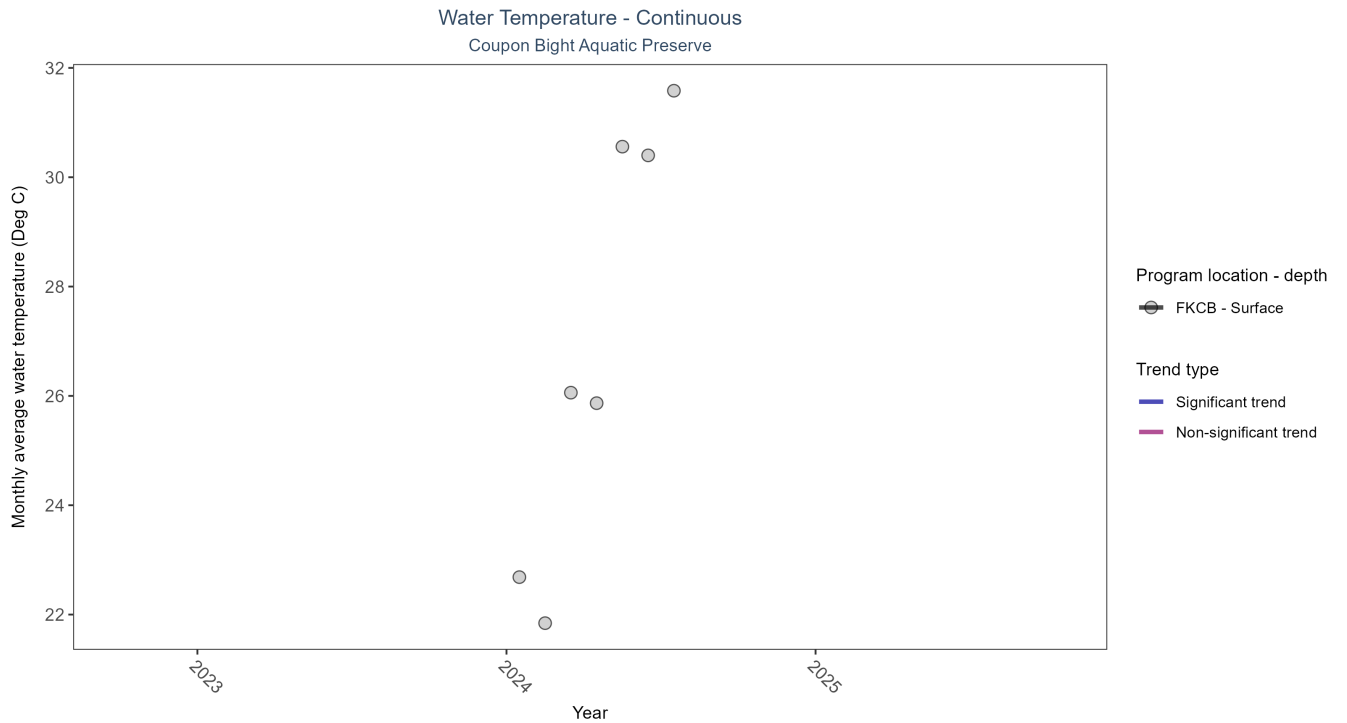


Figure 22: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 27: Seasonal Kendall-Tau Results for Water Temperature - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data to calculate trend	16263	1	2024 - 2024	26.8	-	-	-	-

There was insufficient data to fit a model for one location.

Salinity - Continuous

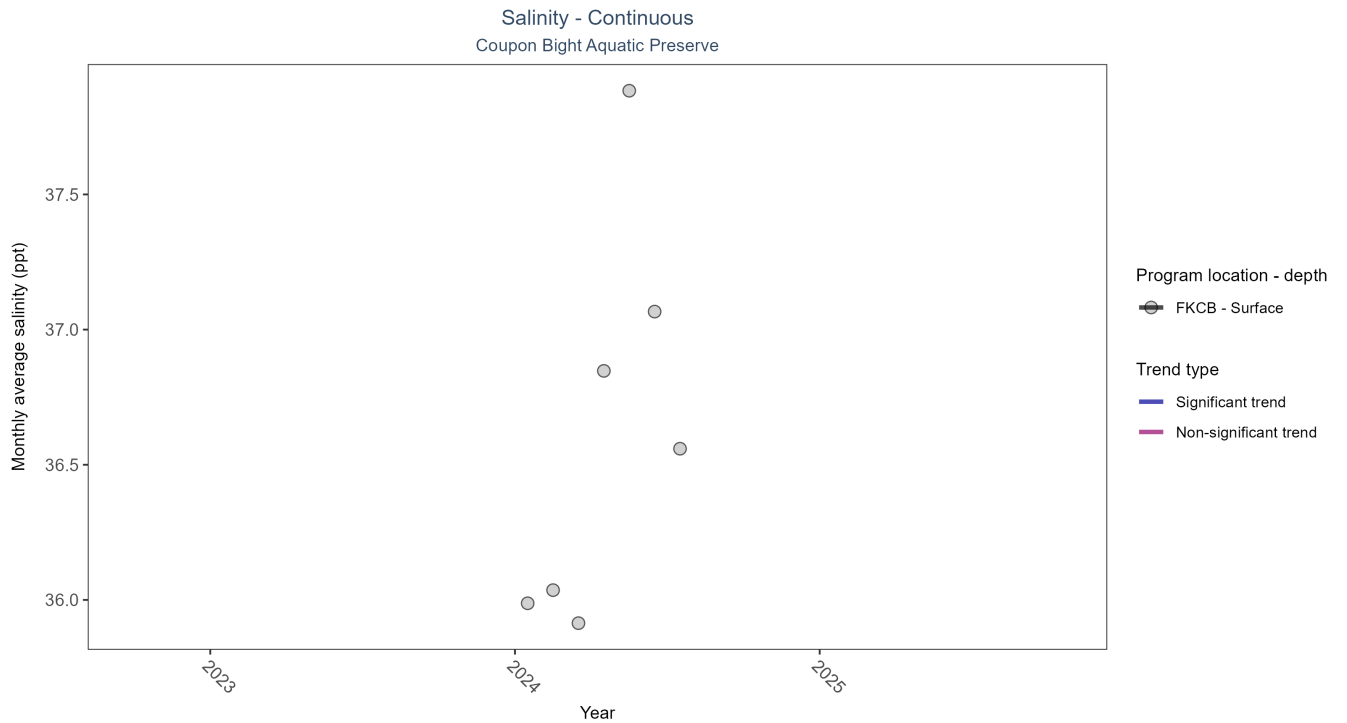


Figure 23: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 28: Seasonal Kendall-Tau Results for Salinity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data to calculate trend	16258	1	2024 - 2024	36.5	-	-	-	-

There was insufficient data to fit a model for one location.

Specific Conductivity - Continuous

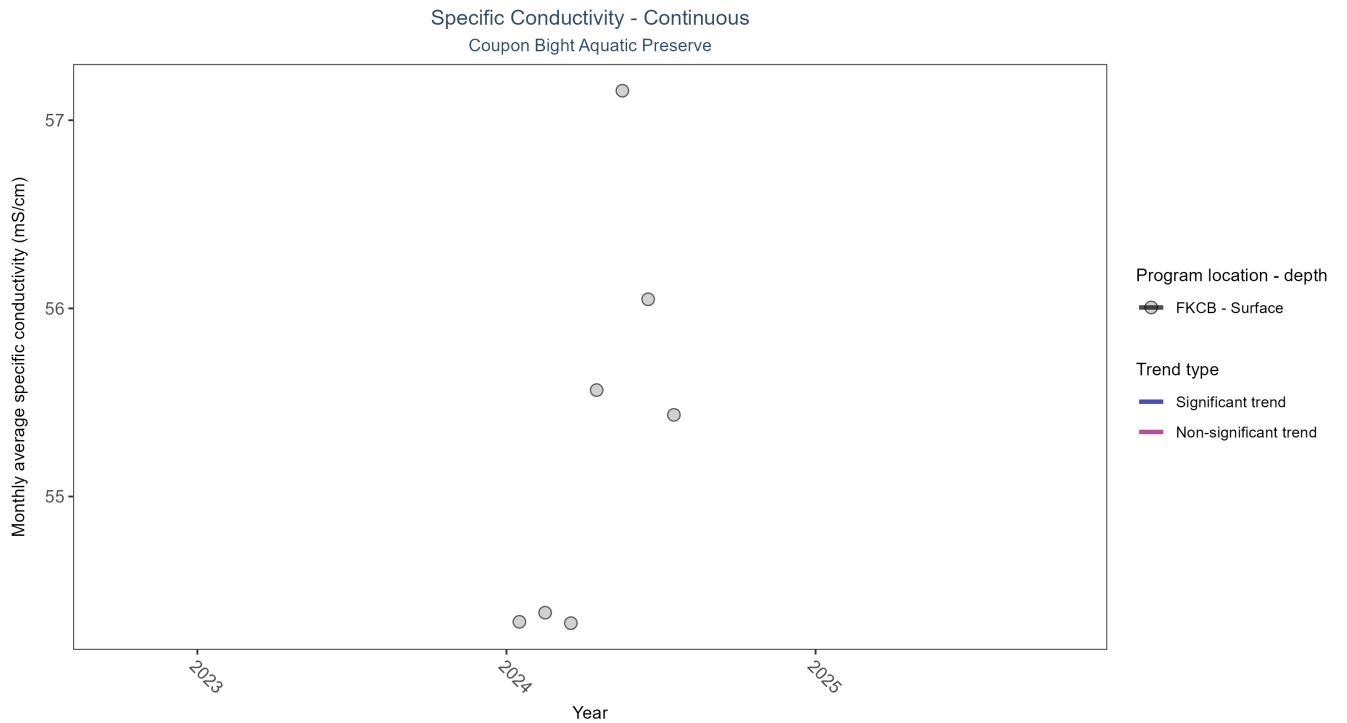


Figure 24: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 29: Seasonal Kendall-Tau Results for Specific Conductivity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data to calculate trend	16258	1	2024 - 2024	55.12	-	-	-	-

There was insufficient data to fit a model for one location.

Coral Reef

The data file used is: **All_CORAL_Parameters-2026-Jun-04.txt**
Species Richness

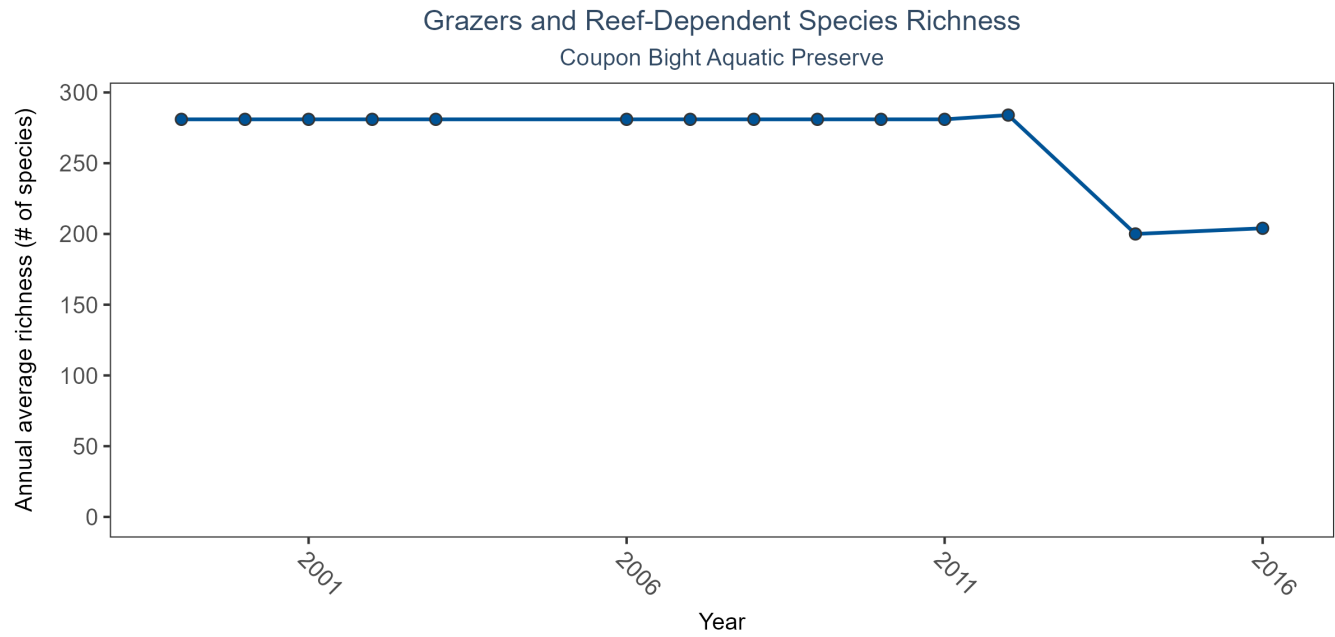


Figure 25: Line graph of annual average species richness of grazers and reef-dependent species over time. If the time series included more than one year of observations, a line connects the data points for visualization.

Table 30: Coral Species Richness

<i>No. of Samples</i>	<i>No. Years with Data</i>	<i>Period of Record</i>	<i>Median No. of Taxa</i>	<i>Mean No. of Taxa</i>
49	14	1999 - 2016	281	266.6735

The median annual number of taxa was 281 based on 49 observations collected between 1999 and 2016.

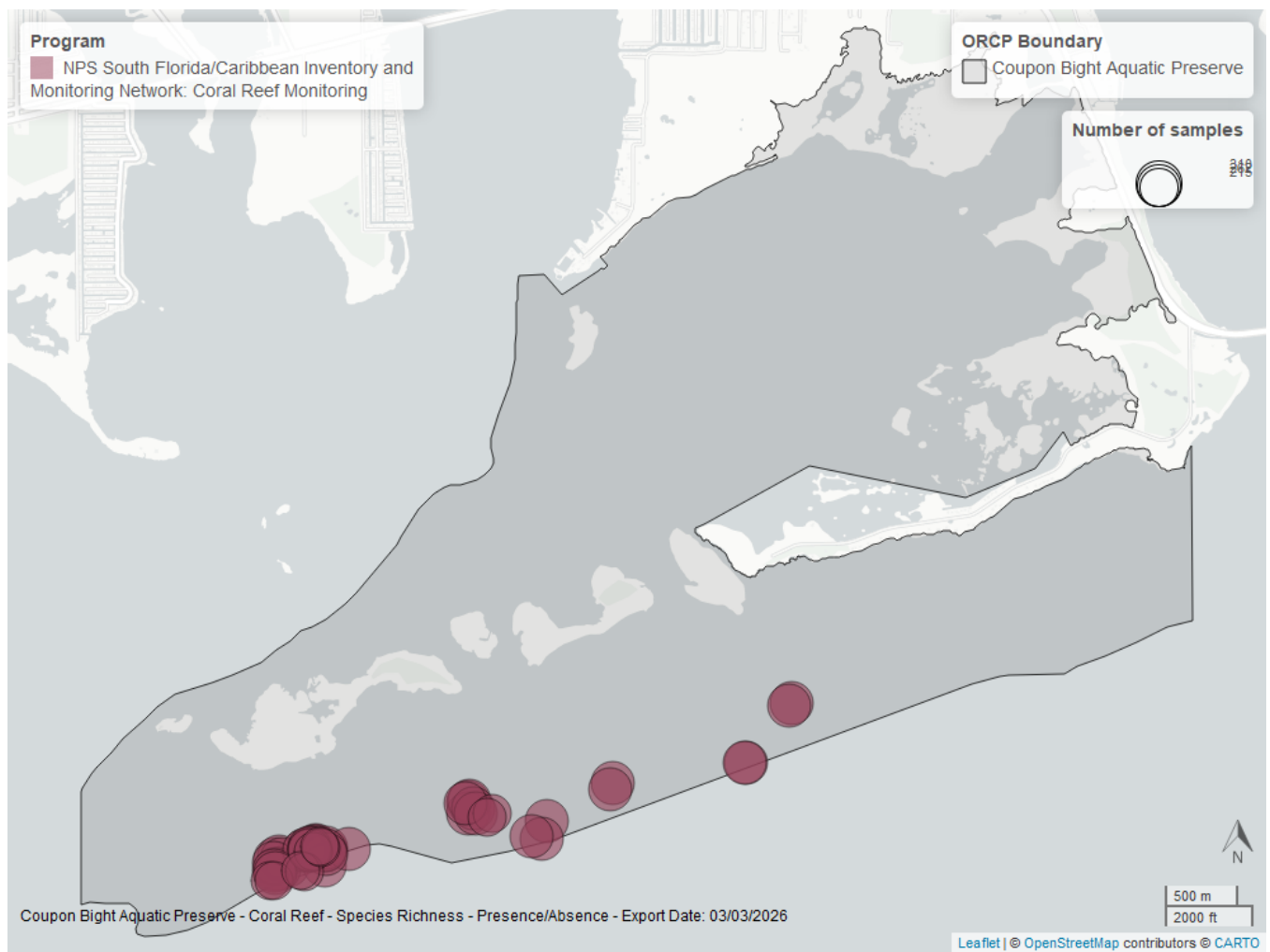


Figure 26: Map showing location of coral species richness sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Species list

<i>Abudefduf saxatilis</i> ¹	<i>Epinephelus drummondhayi</i> ¹	<i>Oligoplites saurus</i> ¹
<i>Acanthemblemaria aspera</i> ¹	<i>Epinephelus guttatus</i> ¹	<i>Ophioblennius macclurei</i> ¹
<i>Acanthemblemaria chaplini</i> ¹	<i>Epinephelus itajara</i> ¹	<i>Opistognathus aurifrons</i>
<i>Acanthemblemaria maria</i> ¹	<i>Epinephelus morio</i> ¹	<i>Opistognathus macrognathus</i>
<i>Acanthemblemaria spinosa</i> ¹	<i>Epinephelus striatus</i> ¹	<i>Opistognathus</i> sp.
<i>Acanthocybium solandri</i>	<i>Eques lanceolatus</i> ¹	<i>Opistognathus whitehursti</i>
<i>Acanthostracion polygonium</i> ¹	<i>Equetus punctatus</i> ¹	<i>Orbicella annularis</i> ²
<i>Acanthostracion quadricornis</i> ¹	<i>Erylus formosus</i>	<i>Orbicella faveolata</i> ²
<i>Acanthurus bahianus</i> ¹	<i>Erythropodium caribaeorum</i> ²	<i>Orbicella franksi</i> ²
<i>Acanthurus chirurgus</i> ¹	<i>Eucinostomus argenteus</i>	<i>Orthopristis chrysoptera</i> ¹
<i>Acanthurus coeruleus</i> ¹	<i>Eucinostomus gula</i>	Other calcareous macroalgae
<i>Acanthurus</i> sp. ¹	<i>Eucinostomus jonesii</i>	Other fleshy macroalgae
<i>Acropora cervicornis</i> ²	<i>Eunicea calyculata</i> ²	Other green algae
<i>Acropora palmata</i> ²	<i>Eunicea flexuosa</i> ²	<i>Oxyurichthys stigmaphius</i> ¹
Actiniaria	<i>Eunicea fusca</i> ²	<i>Pagrus pagrus</i> ¹
<i>Aetobatus narinari</i>	<i>Eunicea knighti</i> ²	<i>Palythoa mammosa</i>
<i>Agaricia agaricites</i> ²	<i>Eunicea laciniata</i> ²	<i>Pandaros acanthifolium</i>
<i>Agaricia fragilis</i> ²	<i>Eunicea laxispica</i> ²	<i>Parablennius marmoreus</i> ¹
<i>Agaricia grahamae</i> ²	<i>Eunicea mammosa</i> ²	<i>Paraclinus marmoratus</i> ¹
<i>Agaricia humilis</i> ²	<i>Eunicea palmeri</i> ²	<i>Paraclinus nigripinnis</i> ¹
<i>Agaricia lamarcki</i> ²	<i>Eunicea succinea</i> ²	<i>Paralichthys albigutta</i> ¹
<i>Agaricia</i> spp.	<i>Eunicea tourneforti</i> ²	<i>Paranthias furcifer</i> ¹
<i>Agaricia undata</i> ²	<i>Eusmilia fastigiata</i> ²	<i>Pareques acuminatus</i> ¹
<i>Agelas clathrodes</i>	<i>Euthynnus alletteratus</i>	<i>Pareques umbrosus</i> ¹
<i>Agelas conifera</i>	<i>Favia fragum</i> ²	<i>Pempheris schomburgkii</i> ¹
<i>Agelas dispar</i>	Fine turf	<i>Penicillus</i> spp.
<i>Agelas schmidtii</i>	<i>Fistularia tabacaria</i> ¹	<i>Peyssonnelia</i>
<i>Agelas wiedenmayeri</i>	<i>Fowlerichthys ocellatus</i> ¹	<i>Phaeoptyx xenus</i> ¹
<i>Ahlia egmontis</i>	<i>Galaxaura</i> spp.	<i>Phorbas</i> sp.
<i>Aiolochoiria crassa</i>	<i>Geodia gibberosa</i>	<i>Phyllangia americana</i> ²
<i>Albula vulpes</i>	<i>Geodia neptuni</i>	<i>Phymanthus crucifer</i>
<i>Alcyonacea</i> sp. ²	<i>Gerres cinereus</i>	<i>Plakortis angulospiculatus</i>
<i>Alectis ciliaris</i> ¹	<i>Ginglymostoma cirratum</i>	<i>Plexaura homomalla</i> ²
<i>Alphestes afer</i> ¹	<i>Gnatholepis thompsoni</i> ¹	<i>Plexaura kuna</i> ²
<i>Aluterus monoceros</i> ¹	<i>Gobioclinus bucciferus</i> ¹	<i>Plexaurella dichotoma</i> ²
<i>Aluterus schoepfii</i> ¹	<i>Gobioclinus filamentosus</i> ¹	<i>Plexaurella grandiflora</i> ²
<i>Aluterus scriptus</i> ¹	<i>Gobioclinus gobicus</i> ¹	<i>Plexaurella grisea</i> ²
<i>Aluterus</i> sp. ¹	<i>Gobioclinus kalisherae</i> ¹	<i>Plexaurella nutans</i> ²
<i>Amblycirrhitus pinos</i> ¹	<i>Gobiosoma</i> sp. ¹	<i>Pomacanthus arcuatus</i> ¹
<i>Amphimedon compressa</i>	<i>Gorgonia flabellum</i> ²	<i>Pomacanthus paru</i> ¹
<i>Amphimedon viridis</i>	<i>Gorgonia mariae</i> ²	Porifera
<i>Amphiroa</i> spp.	<i>Gorgonia ventalina</i> ²	Porifera spp.
<i>Anchoa lyolepis</i>	<i>Gramma loreto</i> ¹	<i>Porites astreoides</i> ²
<i>Anisotremus surinamensis</i> ¹	<i>Gymnothorax funebris</i> ¹	<i>Porites branneri</i> ²
<i>Anisotremus virginicus</i> ¹	<i>Gymnothorax miliaris</i> ¹	<i>Porites colonensis</i> ²
<i>Antillogorgia acerosa</i> ²	<i>Gymnothorax moringa</i> ¹	<i>Porites divaricata</i> ²
<i>Antillogorgia americana</i> ²	<i>Gymnothorax nigromarginatus</i> ¹	<i>Porites furcata</i> ²
<i>Antillogorgia bipinnata</i> ²	<i>Gymnothorax saxicola</i> ¹	<i>Porites porites</i> ²
<i>Antillogorgia kallos</i> ²	<i>Gymnothorax vicinus</i> ¹	<i>Porites</i> spp. ²
<i>Antillogorgia rigida</i> ²	<i>Haemulon album</i> ¹	<i>Priacanthus arenatus</i> ¹
<i>Aplysina archeri</i>	<i>Haemulon aurolineatum</i> ¹	<i>Priolepis hipoliti</i> ¹
<i>Aplysina cauliformis</i>	<i>Haemulon carbonarium</i> ¹	<i>Prionotus ophryas</i> ¹
<i>Aplysina fistularis</i>	<i>Haemulon flavolineatum</i> ¹	<i>Pristipomoides aquilonaris</i> ¹
<i>Aplysina fulva</i>	<i>Haemulon macrostomum</i> ¹	<i>Pristis pectinata</i>

<i>Aplysina lacunosa</i>	<i>Haemulon melanurum</i> ¹	<i>Prognathodes aculeatus</i> ¹
<i>Apogon aurolineatus</i> ¹	<i>Haemulon parra</i> ¹	<i>Pseudobatos lentiginosus</i>
<i>Apogon binotatus</i> ¹	<i>Haemulon plumierii</i> ¹	<i>Pseudodiploria clivosa</i> ²
<i>Apogon maculatus</i> ¹	<i>Haemulon sciurus</i> ¹	<i>Pseudodiploria strigosa</i> ²
<i>Apogon phenax</i> ¹	<i>Haemulon sp.</i> ¹	<i>Pseudoplexaura crucis</i> ²
<i>Apogon pseudomaculatus</i> ¹	<i>Haemulon striatum</i> ¹	<i>Pseudoplexaura flagellosa</i> ²
<i>Apogon quadrisquamatus</i> ¹	<i>Haemulon vittatum</i> ¹	<i>Pseudoplexaura porosa</i> ²
<i>Apogon townsendi</i> ¹	<i>Halichoeres bivittatus</i> ¹	<i>Pseudoplexaura wagneri</i> ²
<i>Archosargus probatocephalus</i> ¹	<i>Halichoeres caudalis</i> ¹	<i>Pseudupeneus maculatus</i> ¹
<i>Archosargus rhomboidalis</i> ¹	<i>Halichoeres cyanocephalus</i> ¹	<i>Ptereleotris calliura</i>
<i>Arturia canariensis</i>	<i>Halichoeres garnoti</i> ¹	<i>Ptereleotris helenae</i>
<i>Astrapogon puncticulatus</i> ¹	<i>Halichoeres maculipinna</i> ¹	<i>Pterogorgia anceps</i> ²
<i>Astrapogon sp.</i> ¹	<i>Halichoeres pictus</i> ¹	<i>Pterogorgia citrina</i> ²
<i>Astrapogon stellatus</i> ¹	<i>Halichoeres poeyi</i> ¹	<i>Pterogorgia guadalupensis</i> ²
<i>Astroscoptes guttatus</i>	<i>Halichoeres radiatus</i> ¹	<i>Pterois miles</i> ¹
<i>Atherinomorus stipes</i>	<i>Haliclona (Reneira) aqueductus</i>	<i>Pterois volitans</i> ¹
<i>Aulostomus maculatus</i> ¹	<i>Haliclona (Reniera) tubifera</i>	<i>Ptilocaulis sp.</i>
<i>Axinellida</i>	<i>Haliclona sp.</i>	<i>Rachycentron canadum</i>
<i>Azurina cyanea</i> ¹	<i>Halimeda spp.</i>	<i>Razorfish sp.</i> ¹
<i>Balistes capricornis</i> ¹	<i>Halisarca sp.</i>	<i>Red calcareous branching algae</i>
<i>Balistes sp.</i> ¹	<i>Harengula humeralis</i>	<i>Red frondose algae</i>
<i>Balistes vetula</i> ¹	<i>Harengula jaguana</i>	<i>Remora remora</i>
<i>Bare substrate</i>	<i>Helioseris cucullata</i> ²	<i>Rhodactis osculifera</i>
<i>Bartholomea annulata</i>	<i>Hemimblemia simula</i> ¹	<i>Rhomboplites aurorubens</i> ¹
<i>Blenniidae sp.</i> ¹	<i>Hemiramphus brasiliensis</i>	<i>Ricordea florida</i>
<i>Bodianus pulchellus</i> ¹	<i>Heteroconger longissimus</i>	<i>Rubble</i>
<i>Bodianus rufus</i> ¹	<i>Heteropriacanthus cruentatus</i> ¹	<i>Rypticus maculatus</i> ¹
<i>Bollmannia boqueronensis</i> ¹	<i>Higginsia strigilata</i>	<i>Rypticus saponaceus</i> ¹
<i>Bothus lunatus</i> ¹	<i>Hippocampus erectus</i> ¹	<i>Sand-sand</i>
<i>Bothus ocellatus</i> ¹	<i>Hippospongia sp.</i>	<i>Sand on hard-bottom</i>
<i>Brachygenys chrysargyreus</i> ¹	<i>Holacanthus bermudensis</i> ¹	<i>Sardinella aurita</i>
<i>Branching gorgonian</i> ²	<i>Holacanthus ciliaris</i> ¹	<i>Sargassum spp.</i>
<i>Briareum asbestinum</i> ²	<i>Holacanthus tricolor</i> ¹	<i>Sargocentron coruscum</i> ¹
<i>Brockius nigricinctus</i> ¹	<i>Holocentrus adscensionis</i> ¹	<i>Sargocentron vexillarium</i> ¹
<i>Brotula barbata</i>	<i>Holocentrus rufus</i> ¹	<i>Scartella cristata</i> ¹
<i>Brown algae</i>	<i>Hypanus americanus</i>	<i>Scarus coelestinus</i> ¹
<i>Bryozoa</i>	<i>Hypoleurochilus bermudensis</i> ¹	<i>Scarus coerules</i> ¹
<i>Calamus bajonado</i> ¹	<i>Hypoatherina harringtonensis</i>	<i>Scarus guacamaia</i> ¹
<i>Calamus calamus</i> ¹	<i>Hypoplectrus chlorurus</i> ¹	<i>Scarus iseri</i> ¹
<i>Calamus nodosus</i> ¹	<i>Hypoplectrus gemma</i> ¹	<i>Scarus sp.</i> ¹
<i>Calamus penna</i> ¹	<i>Hypoplectrus guttarius</i> ¹	<i>Scarus taeniopterus</i> ¹
<i>Calamus proridens</i> ¹	<i>Hypoplectrus hybrid</i> ¹	<i>Scarus vetula</i> ¹
<i>Calcareous green algae</i>	<i>Hypoplectrus indigo</i> ¹	<i>Schultzea beta</i> ¹
<i>Callionymus bairdi</i> ¹	<i>Hypoplectrus nigricans</i> ¹	<i>Scleractinia</i> ²
<i>Callyspongia (Callyspongia) fallax</i>	<i>Hypoplectrus puella</i> ¹	<i>Scolymia sp.</i>
<i>Callyspongia (Cladochalina) aculeata</i>	<i>Hypoplectrus sp.</i> ¹	<i>Scolymia spp.</i> ²
<i>Callyspongia (Cladochalina) plicifera</i>	<i>Hypoplectrus tann</i> ¹	<i>Scomberomorus cavalla</i>
<i>Callyspongia (Cladochalina) tenerima</i>	<i>Hypoplectrus unicolor</i> ¹	<i>Scomberomorus maculatus</i>
<i>Calyx podatypa</i>	<i>Hyporthodus flavolimbatus</i> ¹	<i>Scomberomorus regalis</i>
<i>Cantherhines macrocerus</i> ¹	<i>Hyporthodus niveatus</i> ¹	<i>Scopalina ruetzleri</i>
<i>Cantherhines pullus</i> ¹	<i>Hyrtios violaceus</i>	<i>Scorpaena plumieri</i> ¹
<i>Canthidermis sufflamen</i> ¹	<i>Iciligorgia schrammi</i> ²	<i>Scorpaenodes caribbaeus</i> ¹
<i>Canthigaster rostrata</i> ¹	<i>Iotrochota birotulata</i>	<i>Selachii</i>
<i>Caranx bartholomaei</i> ¹	<i>Ircinia campana</i>	<i>Selene vomer</i> ¹
<i>Caranx crysos</i> ¹	<i>Ircinia felix</i>	<i>Seriola dumerili</i> ¹
<i>Caranx hippos</i> ¹	<i>Ircinia strobilina</i>	<i>Seriola rivoliana</i> ¹
<i>Caranx latus</i> ¹	<i>Ircinia variabilis</i>	<i>Seriola sp.</i> ¹

<i>Caranx lugubris</i> ¹	<i>Isophyllia rigida</i> ²	<i>Serranid</i> sp. ¹
<i>Caranx ruber</i> ¹	<i>Isophyllia sinuosa</i> ²	<i>Serranus annularis</i> ¹
<i>Caranx</i> sp. ¹	<i>Istiophorus platypterus</i>	<i>Serranus baldwini</i> ¹
<i>Carcharhinus falciformis</i>	<i>Jania</i> spp.	<i>Serranus phoebe</i> ¹
<i>Carcharhinus leucas</i>	<i>Jenkinsia</i> sp.	<i>Serranus subligarius</i> ¹
<i>Carcharhinus limbatus</i>	<i>Kallymenia</i> spp.	<i>Serranus tabacarius</i> ¹
<i>Carcharhinus perezii</i>	<i>Kyphosus sectatrix</i> ¹	<i>Serranus tigrinus</i> ¹
<i>Centropomus undecimalis</i>	<i>Labrisomidae</i> sp. ¹	<i>Serranus tortugarum</i> ¹
<i>Centropyge argi</i> ¹	<i>Labrisomus nuchipinnis</i> ¹	<i>Siderastrea radians</i> ²
<i>Cephalopholis cruentata</i> ¹	<i>Lachnolaimus maximus</i> ¹	<i>Siderastrea siderea</i> ²
<i>Cephalopholis fulva</i> ¹	<i>Lactophrys bicaudalis</i> ¹	Silt on hard-bottom
<i>Chaenopsis limbaughii</i> ¹	<i>Lactophrys trigonus</i> ¹	<i>Siphonodictyon coralliphagum</i>
<i>Chaetodipterus faber</i>	<i>Lactophrys triqueter</i> ¹	<i>Siphonodictyon siphonum</i>
<i>Chaetodon capistratus</i> ¹	<i>Lagodon rhomboides</i> ¹	<i>Snapper</i> sp. ¹
<i>Chaetodon ocellatus</i> ¹	<i>Laurencia</i> spp.	<i>Solenastrea bournoni</i> ²
<i>Chaetodon sedentarius</i> ¹	<i>Lebrunia neglecta</i>	<i>Solenastrea hyades</i> ²
<i>Chaetodon striatus</i> ¹	<i>Liagora</i> spp.	<i>Sparidae</i> sp. ¹
<i>Chilomycterus antennatus</i> ¹	<i>Liopropoma eukrines</i> ¹	<i>Sparisoma atomarium</i> ¹
<i>Chilomycterus reticulatus</i> ¹	<i>Liopropoma mowbrayi</i> ¹	<i>Sparisoma aurofrenatum</i> ¹
<i>Chilomycterus schoepfii</i> ¹	<i>Liopropoma rubre</i> ¹	<i>Sparisoma chrysopterum</i> ¹
<i>Chloroscombrus chrysurus</i> ¹	<i>Lobophora</i> spp.	<i>Sparisoma radians</i> ¹
<i>Chondrilla nucula</i>	<i>Lutjanus analis</i> ¹	<i>Sparisoma rubripinne</i> ¹
<i>Chondrosia</i> sp.	<i>Lutjanus apodus</i> ¹	<i>Sparisoma</i> sp. ¹
<i>Chriodorus atherinoides</i>	<i>Lutjanus buccanella</i> ¹	<i>Sparisoma viride</i> ¹
<i>Chromis enchrysurus</i> ¹	<i>Lutjanus cyanopterus</i> ¹	<i>Spheciospongia vesparium</i>
<i>Chromis insolata</i> ¹	<i>Lutjanus griseus</i> ¹	<i>Sphoeroides</i> ¹
<i>Chromis multilineata</i> ¹	<i>Lutjanus jocu</i> ¹	<i>Sphoeroides spengleri</i> ¹
<i>Chromis scotti</i> ¹	<i>Lutjanus mahogoni</i> ¹	<i>Sphoeroides testudineus</i> ¹
<i>Cinachya</i> sp.	<i>Lutjanus synagris</i> ¹	<i>Sphyraena barracuda</i> ¹
<i>Cladocora arbuscula</i> ²	<i>Madracis carmabi</i> ²	<i>Sphyraena guachancho</i>
<i>Clathria</i> (Thalysias) <i>venosa</i>	<i>Madracis decactis</i> ²	<i>Sphyraena picudilla</i>
<i>Clathria</i> (Thalysias) <i>virgultosa</i>	<i>Madracis formosa</i> ²	<i>Sphyrna lewini</i>
<i>Clathria</i> sp.	<i>Madracis myriaster</i> ²	<i>Sphyrna mokarran</i>
<i>Clepticus parrae</i> ¹	<i>Madracis senaria</i> ²	<i>Sphyrna tiburo</i>
<i>Cliona caribbaea</i>	<i>Malacanthus plumieri</i>	<i>Spirastrella coccinea</i>
<i>Cliona delitrix</i>	<i>Malacoctenus aurolineatus</i> ¹	<i>Spirastrella mollis</i>
<i>Cliona</i> sp.	<i>Malacoctenus gilli</i> ¹	<i>Spongia</i> sp.
<i>Cliona</i> spp.	<i>Malacoctenus macropus</i> ¹	<i>Squirrelfish</i> sp. ¹
<i>Cliona varians</i>	<i>Malacoctenus triangulatus</i> ¹	<i>Stegastes adustus</i> ¹
<i>Colpophyllia natans</i> ²	<i>Malacoctenus versicolor</i> ¹	<i>Stegastes dienaecus</i> ¹
<i>Condylactis gigantea</i>	<i>Manicina areolata</i> ²	<i>Stegastes leucostictus</i> ¹
<i>Corallimorpharians</i>	<i>Meandrina meandrites</i> ²	<i>Stegastes partitus</i> ¹
<i>Coryphopterus dicrus</i> ¹	<i>Megalops atlanticus</i>	<i>Stegastes planifrons</i> ¹
<i>Coryphopterus eidolon</i> ¹	<i>Melichthys niger</i> ¹	<i>Stegastes</i> sp. ¹
<i>Coryphopterus glaucofraenum</i> ¹	<i>Menidia</i> sp.	<i>Stegastes variabilis</i> ¹
<i>Coryphopterus lipernes</i> ¹	<i>Microgobius carri</i> ¹	<i>Stephanocoenia intersepta</i> ²
<i>Coryphopterus personatus</i> ¹	<i>Microgobius microlepis</i> ¹	<i>Stephanolepis hispidia</i> ¹
<i>Coryphopterus punctipectophorus</i> ¹	<i>Microspathodon chrysurus</i> ¹	<i>Stichodactyla helianthus</i>
<i>Coryphopterus</i> sp. ¹	<i>Millepora alcicornis</i> ²	<i>Strongylacidon</i> sp.
<i>Cribrochalina vasculum</i>	<i>Millepora complanata</i> ²	<i>Strongylura notata</i> ¹
<i>Crustose coralline algae</i>	<i>Mobula birostris</i>	<i>Strongylura timucu</i>
<i>Cryptotomus roseus</i> ¹	<i>Monacanthus ciliatus</i> ¹	<i>Stygnobrotula latebricola</i>
<i>Ctenogobius saepepallens</i> ¹	<i>Monacanthus tuckeri</i> ¹	<i>Stypopodium</i> spp.
<i>Cyanobacteria</i>	<i>Monanchora arbuscula</i>	<i>Syacium micrurum</i> ¹
<i>Dactylopterus volitans</i>	<i>Montastraea cavernosa</i> ²	<i>Syngnathus scovelli</i> ¹
<i>Decapterus macarellus</i> ¹	<i>Mulloidichthys martinicus</i> ¹	<i>Synodus foetens</i> ¹
<i>Decapterus punctatus</i> ¹	<i>Muraena retifera</i> ¹	<i>Synodus intermedius</i> ¹

Decapterus sp. ¹	Muricea atlantica ²	Synodus synodus ¹
Dendrogyra cylindrus ²	Muricea elongata ²	Tectitethya crypta
Dermatolepis inermis ¹	Muricea laxa ²	Tedania (Tedania) ignis
Desmopsamma anchorata	Muricea muricata ²	Tethya diploderma
Diadema antillarum	Muricea pinnata ²	Thalassoma bifasciatum ¹
Dichocoenia stokesii ²	Muriceopsis flavida ²	Thick turf
Dictyota spp.	Mussa angulosa ²	Tigriobius macrodon ¹
Diodon holocanthus ¹	Mycale (Mycale) laevis	Tigriobius saucrus ¹
Diodon hystrix ¹	Mycale sp.	Trachinotus falcatus ¹
Diodon sp. ¹	Mycetophyllia aliciae ²	Trachyteleia hispida
Diplastrella megastellata	Mycetophyllia danaana ²	Tunicata
Diplectrum formosum ¹	Mycetophyllia ferox ²	Turf algae free of sediment
Diplodus argenteus ¹	Mycetophyllia lamarckiana ²	Turf algae with sediment
Diplodus holbrookii ¹	Mycetophyllia spp. ²	Tylosurus crocodilus
Diploria labyrinthiformis ²	Mycteroperca acutirostris ¹	Udotea spp.
Discosoma carlgreni	Mycteroperca bonaci ¹	Umbrina coroides ¹
Doratonotus megalepis ¹	Mycteroperca interstitialis ¹	Unidentified species
Dragmacidon lunaecharta	Mycteroperca microlepis ¹	Unknown black smooth encrusting sponge
Dysidea etheria	Mycteroperca phenax ¹	Unknown bowling ball sponge
Dysidea fragilis	Mycteroperca tigris ¹	Unknown brown encrusting sponge
Dysidea janiae	Mycteroperca venenosa ¹	Unknown brown smooth sponge
Echeneis naucrates	Myrichthys breviceps	Unknown brown tube sponge
Echeneis neucratoides	Myrichthys ocellatus	Unknown brown vein sponge
Ectyoplasia ferox	Myripristis jacobus ¹	Unknown green encrusting sponge
Elacatinus dilepis ¹	Narcine bancroftii	Unknown olive sponge
Elacatinus evelynae ¹	Needlefish sp.	Unknown orange encrusting sponge
Elacatinus horsti ¹	Negaprion brevirostris	Unknown orange massive sponge
Elacatinus oceanops ¹	Neofibularia nolitangere	Unknown pink lumpy sponge
Elacatinus randalli ¹	Neoniphon marianus ¹	Unknown red encrusting sponge
Elacatinus xanthiprora ¹	Neopetrosia carbonaria	Unknown red lumpy tube sponge
Elagatis bipinnulata ¹	Nes longus ¹	Unknown red squishy sponge
Elops saurus	Nicholsina usta ¹	Urobatis jamaicensis
Emblemaria pandionis ¹	Niphates amorpha	Verongula gigantea
Emblemariopsis bahamensis ¹	Niphates digitalis	Verongula reiswigi
Emmelichthyops atlanticus ¹	Niphates erecta	Verongula rigida
Enchelycore nigricans ¹	Oculina diffusa ²	Xestospongia muta
Encrusting gorgonian ²	Oculina sp.	Xyrichtys martinicensis ¹
Enneanectes altivelis	Ocyurus chrysurus ¹	Xyrichtys novacula ¹
Enneanectes boehlkei	Odontoscion dentex ¹	Xyrichtys splendens ¹
Epinephelus adscensionis ¹	Ogocephalus sp.	Zoanthids

1 - Coral Reef - Species Richness, 2 - Coral Reef - Percent Cover

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